Phonetic and Phonological Sound Changes in an Agent-Based Model

Michele Gubian*[1], Johanna Cronenberg[1], and Jonathan Harrington[1]

[1] *Institute of Phonetics and Speech Processing, Ludwig Maximilian University of Munich*

Abstract

The focus of the study is an agent-based model (ABM) concerned with simulating phonological stability and change using real speech data from a population of speakers. At the core of the model was a flexible and agent-specific association between acoustic exemplars and phonological categories. This was achieved by means of two general-purpose unsupervised machine learning algorithms: the first grouped exemplars into acoustic clusters, the second identified sets of clusters which largely contain exemplars of the same word classes. The model was tested first on data from Standard Southern British English, where a shift of /u/ to the front of the vowel space was expected not to cause any phonological re-categorisation. The simulation indeed showed phonological stability despite the phonetic change. Using the same settings, the ABM was then applied to data from New Zealand English in which a merger of /eə, ɪə/ diphthongs has taken place in the last 50 years. Compatibly with that, the simulation showed a shift from /eə/ towards /ɪə/ along with the neutralisation of the diphthong contrast. The results are discussed with respect to the conditions necessary to model a merger, taking into account both findings on marginal phonological relationships and principles from exemplar theory.

Keywords— Sound Change, Agent-Based Modelling, Gaussian Mixture Models, Non-Negative Matrix Factorisation, New Zealand English

1 Introduction

The present study is about the development of a cognitively-inspired computational model, the interactive phonetic (IP) model of sound change (Harrington et al., 2018; Harrington & Schiel, 2017; Stevens et al., 2019). Compatibly with many other computational models of sound change (Blevins & Wedel, 2009; Kirby, 2014; Sóskuthy, 2015; Stanford & Kenny, 2013; Todd et al., 2019; Wedel, 2006, 2012), the IP model has a particular focus on explaining the emergence and spread of sound change, given plausible hypotheses about both the relationship between speech production and perception and the cognitive architecture that links the speech signal, some form of abstract phonological code, and the lexicon. The approach in the IP model departs methodologically from most of the other models of sound change in two ways. Firstly, the input to our model is not artificially constructed but is instead real speech data recorded from actual speakers. That is, each speaker is represented by an interacting computational agent initialised with speech signals and a lexicon. The purpose of doing so has been to ground the computational model in phonetic variation; and also because this approach allows an exploration of how individual speaker characteristics, idiosyncrasies, and innovations are related to population change (Baker et al., 2011; François, 2014; Garrett & Johnson, 2011; Hall-Lew et al., 2021; Samuels, 1972; Yu, 2021; Yu & Zellou, 2019). Secondly, the input to the IP model typically consists of parameters that capture characteristics of the time-varying

*Corresponding author at: Institute of Phonetics and Speech Processing, Ludwig Maximilian University of Munich, Schellingstr. 3 (VG, room 224), D-80799 Munich.
E-mail address: m.gubian@lmu.de

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speech signal instead of those that describe a static snapshot of speech (such as the temporal midpoint of vowels). The reason for this is because almost all of regular sound change originates from processes that are inherently dynamic such as coarticulation (Beddor, 2009; Garrett & Johnson, 2011; Lindblom et al., 1995; Ohala, 1993), undershoot (Beckman et al., 1992; Lindblom et al., 1995), or associations between events that are distributed in time that give rise to metathesis (Blevins & Garrett, 2004) and dissimilation (Abrego-Collier, 2013; Alderete & Frisch, 2007).

In this work, the aim is to extend the IP model such that it can also handle sound changes with a phonological component, i.e. where exemplars and words are flexibly linked to phonological classes. In particular, we are interested in modelling mergers which is when two phonological classes become acoustically inseparable and the phonological contrast is finally neutralised. The mechanisms that are required from a modelling perspective in order to be able to handle this kind of sound change are considered in section 2 before the IP model and the revisions that have been made to include the proposed mechanisms are introduced. Two tests of the revised model follow: The first, in section 3, is on /u/-fronting in Standard Southern British English (SSBE), a sound change that was the subject of a previous study by Harrington and Schiel (2017). This study was chosen for replication because only a phonetic shift but no phonological changes were expected and the revised IP model should still be capable of modelling phonological stability. The second test of the model in section 4 is on the merger of /ɪə/ and /eə/ in New Zealand English (NZE). In this case, we expect both a phonetic shift from /eə/ towards /ɪə/ and a neutralisation of the phonological contrast.

2 Agent-based Model

The agent-based IP model presented in this work is an evolution of the one first introduced in Harrington and Schiel (2017), which was then further modified in Harrington et al. (2018) and Stevens et al. (2019). We describe briefly in section 2.1 the components of this model and then in section 2.2 the main modifications that have been made in order to be able to simulate both the SSBE phonetic shift of /u/-fronting and the NZE merger of /ɪə/ and /eə/. That is, the architecture needs to be capable of modelling phonetic change both with and without additional phonological changes. Importantly, this should not be achieved by a mechanism that can be switched on and off depending on the expected type of change, but instead by using the same model settings independently of the input data and expected outcome.[1]

2.1 The IP model

An agent-based model (ABM) is defined by a set of artificial agents and by the rules and mechanisms by which they interact. In the IP model, agents are capable of producing, perceiving, memorising, and forgetting acoustic exemplars. To do so, they are endowed with a memory, with methods to represent and encode exemplars internally, with rules to determine when to memorise and when to forget exemplars, and with methods to generate new exemplars. Sound change may or may not emerge as a result of the prolonged interactions of the agents and their production-perception loop.

One of the differences of the IP model compared with almost all other models of sound change is that the agents are initialised with speech data from actual recorded speakers. That is, the agents in the IP model are initialised with exemplars which are a data-reduced acoustic parametrisation of the speech sounds under investigation. These exemplars have a fixed association to word classes. There is also a phonological level that links exemplars and words. In earlier versions of the IP model, the phonological classes were either invariable throughout the simulations (Harrington & Schiel, 2017) or regularly updated by performing binary splits or mergers, starting from canonical phonemes that were assigned *a priori* to the exemplars (Stevens et al., 2019). In addition to having to supply phonological labels to begin with, the algorithm in Stevens et al. (2019) tended to result in an unreasonable number of sub-phonemic classes due to a preference for splits. The main innovation in the revised IP model is that the phonological level consists of sub-phonemic classes that are learned in a fully unsupervised fashion (see section 2.2.1).

[1] The code base of the ABM used for this study is available at https://github.com/IPS-LMU/soundChangeR/releases/tag/CompSpeechLang2022. The data is available upon request.

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The interaction rules are fairly simple: agents either freely and randomly interact with each other, or they do so only within or only across predefined groups (e.g., younger vs. older speakers). The agent speaker randomly chooses a word class, builds a Gaussian model over all memorised exemplars associated with that word, and samples a new exemplar from it. Contrary to Todd et al. (2019) and others, the produced tokens are not manipulated by a bias that pushes them into the direction of change. Rather, the conditions for the emergence of systematic phonetic variation are set by the introduction of agent groups initialised with speech representing different stages of an observed sound change. The agent listener receives the exemplar together with the word class (i.e., it is assumed that word recognition works perfectly) and makes use of agent-specific phonological knowledge in order to decide whether or not to memorise the exemplar. There are two memorisation criteria which were variably used in earlier versions of the IP model. The first is maximum *a posteriori* (MAP) probability: an exemplar is memorised only if its posterior probability conditioned on the corresponding phonological class is higher than all other posterior probabilities conditioned on the competing phonological classes.[2] Since all phonological classes are taken into account, this criterion is *relative*. A consequence of using posterior probabilities in this way is that phonological classes repel each other because acoustically ambiguous exemplars are usually rejected by the agent (Liljencrants & Lindblom, 1972; Maclagan & Hay, 2007; Stevens et al., 2019; Wedel, 2006).

The second memorisation criterion is *absolute*, i.e. an exemplar is only memorised if its Mahalanobis distance to the centroid of its corresponding phonological class is lower than a given threshold. In less technical terms, this means that exemplars are rejected if they are atypical members of a phonological class. A possible consequence of the application of this criterion is that asymmetries in the distributions of phonological classes can be amplified, leading to a phonetic shift of a skewed phonological class towards a more compact one. This is because a skewed, spread out phonological class is more likely to integrate new exemplars at its edges than a tightly clustered class. In the revised IP model, both the relative maximum posterior probability and absolute Mahalanobis distance threshold are used as memorisation criteria. An exemplar has to pass both tests in order to be incorporated into the agent listener's memory. This is in line with other models of sound change such as Todd et al. (2019) who apply similar memorisation criteria.

2.2 The revised IP model

Here we consider from the perspective of a cognitively inspired computational model the necessary conditions for a phonological merger. Many computational models (De Boer, 2001; Kirby, 2013, 2014; Sóskuthy, 2015; Stanford & Kenny, 2013; Todd et al., 2019; Wedel, 2006), including the IP model (Harrington et al., 2019; Harrington & Schiel, 2017; Stevens et al., 2019), make use of a perceptual bias against ambiguous tokens. That is, when there are two or more competing phonological categories in the simulated acoustic space, tokens that fall in the ambiguous space between them are less likely to be memorised. This perceptual pressure for contrast maintenance effectively prohibits mergers from happening in these models because phonemes will always repel each other. While the assumption that acoustically close phoneme categories do not always collapse is generally reasonable (Blevins & Wedel, 2009), a holistic model of sound change should be capable of producing both stable phonemic conditions as well as mergers. In order to allow categories to merge, agents must be able to update their association between acoustic tokens, phonemic categories, and lexical classes regularly so as to weaken the effects of the contrast maintenance rule.

In the revised IP model, phonological categories are determined by using both bottom-up acoustic and top-down lexical information. The two-tiered architecture used in this flexible phonology module is explained in section 2.2.1. Section 2.2.2 is about the relation between classical phonemes and the sub-phonemic classes which autonomously emerge from the revised architecture of the IP model. Finally, section 2.2.3 is concerned with solutions to some practical issues regarding the number of exemplars stored in the agents' memories.

2.2.1 Derivation of Phonological Knowledge

Agents store exemplars in their memory. An exemplar contains an acoustic part, which is a data-reduced, usually vectorial representation of a speech sound, and a lexical part, indicating which word the sound belongs to. In the revised IP model, an exemplar is also associated to a sub-phonemic class. This association is not determined *a*

[2] In Harrington and Schiel (2017), a threshold of $\frac{1}{\text{nr. of phonemes}}$ was used instead of MAP probability.

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priori; rather it is periodically computed by an agent on the basis of its memory content. Sub-phonemic classes emerge as a result of applying two different algorithms, each one having a different, partial view on the stored exemplars. The first algorithm groups exemplars into acoustic clusters using only the acoustic information. This is implemented with Gaussian Mixture Models (GMMs, Reynolds, 2009). The second algorithm combines the lexical information with the GMM-derived cluster membership in order to identify sub-phonemic classes which are groupings of acoustic clusters that are formed by a distinct subset of words. More specifically, the second algorithm uses information about the exemplars' assignment to acoustic clusters and about their association to word classes, but not about the location of each exemplar or acoustic cluster in the acoustic space. This second phase is implemented with Non-negative Matrix Factorisation (NMF, Seung & Lee, 2001).

To illustrate, Figure 1 shows acoustic clusters (1a) and sub-phonemic classes (1b) at simulation start from one of the agents from the SSBE case (section 3). In Figure 1a, each dot indicates the position of a memorised exemplar in the acoustic space, and the five ellipses $a_1 - a_5$ represent GMM-derived acoustic clusters. In Figure 1b, the same exemplars are represented by the word class with which they are associated, while colours show the association of exemplars and acoustic clusters (ellipses) to three distinct sub-phonemic classes. For example, because the acoustically close clusters a_2 and a_5 contain the same set of words (*feed*, *heed*, *keyed*, and *seep*), they come to be associated with the same sub-phonemic class (in yellow). On the other hand, even though clusters a_3 and a_4 are acoustically close to each other, they do not form one sub-phonemic class because they contain different words. As it is highly improbable that purely acoustics-based clusters can be grouped precisely according to the words they contain, a tolerance is introduced based on *purity*, i.e. the fraction of exemplars in a cluster or cluster group belonging to a designated set of words. The formation of a sub-phonemic class is regulated by a threshold on purity, i.e. a cluster group needs to contain exemplars of a given set of words with a purity that exceeds the threshold in order for those words to qualify for defining a sub-phonemic class. For example, in Figure 1b two exemplars of *cooed* that belong to a_3 acoustically (in darkred, surrounded by blue words) are 'impurities', as they are members of the sub-phonemic class based on cluster a_4.

Appendices A and B provide more background on GMM and NMF, respectively, while an extended, detailed description of the implementation of the acoustic and sub-phonological levels is provided in Appendix C.

2.2.2 Sub-Phonemes and Their Relation to Classical Phonemes

We consider briefly the relationship between the sub-phonemic classes created by the agents during the ABM simulations and the classical phoneme (or s-phoneme using the terminology in Kiparsky, 2018) and its phonetic variants. There are three main cases to consider.

1. *Phonetic variants of different (classical) phonemes.* These are likely to fall into different sub-phonemic classes using the methodology in Figure 1. This is because they map to mostly different sets of words (e.g. /i/ as in *feed*, *heed*, *weed* vs. /u/ as in *food*, *who'd*, *wooed*) and because they (in this case: /i/ vs. /u/) are acoustically distinct (and so likely to be in different acoustic clusters).

2. *Variants of the same classical phoneme that are in complementary distribution.* These are also likely to fall into different sub-phonemic classes in the ABM, even though they are not contrastive in the classical phonemic sense. Consider for example clear /l/ and dark /ɫ/ in many varieties of English. There are two reasons why these would in most cases be in different sub-phonemic classes. Firstly, they map to different sets of words (clear /l/ mostly to domain-initial position in e.g., *leaf*, *lot*, *luck*; dark /ɫ/ mostly to domain-final position in e.g., *call*, *full*, *sell*). Secondly, clear and dark /l/ have quite different acoustic characteristics (Kirkham, 2017; Lehiste, 1964; Sproat & Fujimura, 1993).

3. *Variants of the same classical phoneme that occur in the same context.* These include free variants (Reinisch et al., 2020): for example, the same speaker might produce word-final /t/ in *bat*, *let*, *sit* sometimes with a strong release/frication and sometimes with hardly any release and glottalisation (Penney et al., 2018; Seyfarth & Garellek, 2020). These variants are acoustically quite different and as a result will tend to fall in two different acoustic clusters. But because the clusters contain predominantly the same sets of words (with /t/ realised with frication vs. glottalisation respectively) they are likely to be grouped into the same sub-phonemic class by an agent. This is the main case in which there is genuine abstraction in the IP model, i.e. in which acoustically quite different variants are mapped to the same sub-phonemic class.

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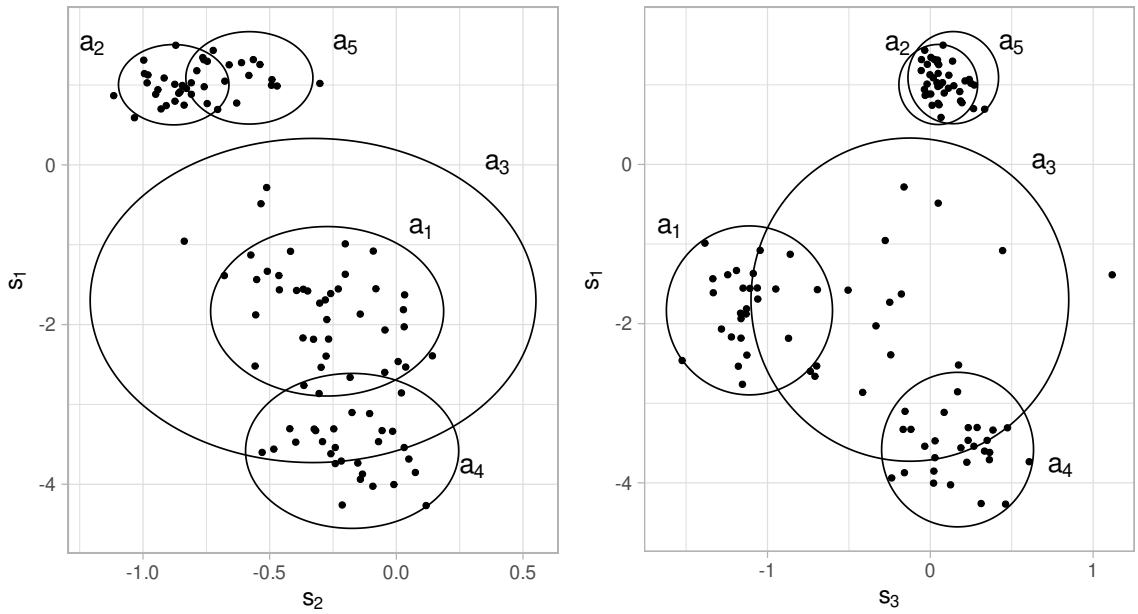
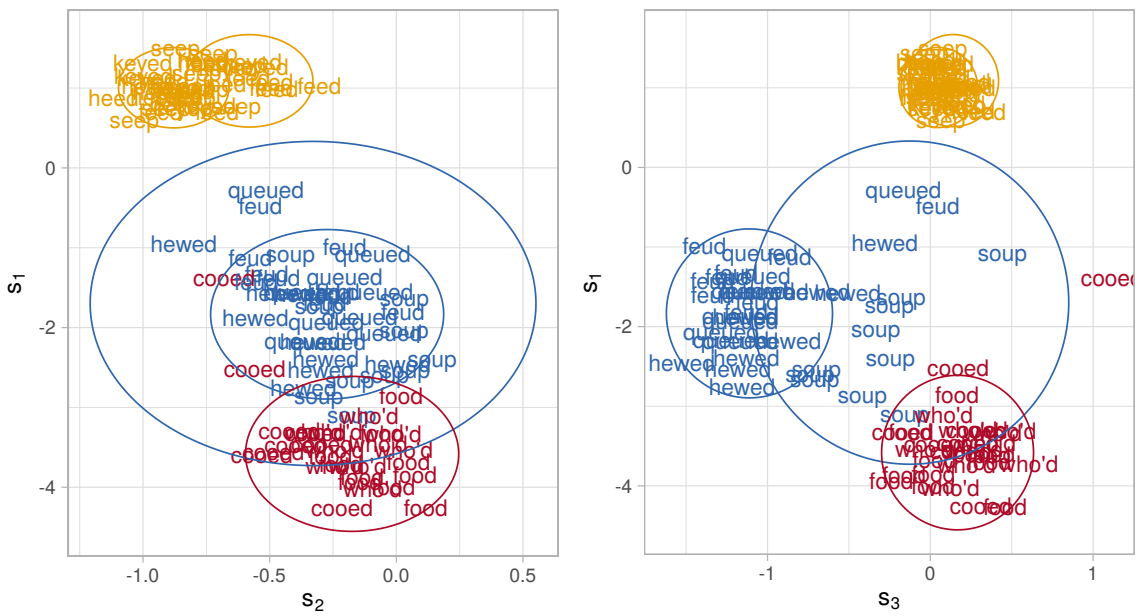


(a)



(b)

Figure 1: Example of acoustic clusters (1a) and sub-phonemic classes (1b) in an agent. In 1a, dots represent memorised exemplars in the acoustic space, and ellipses $a_1 - a_5$ identify acoustic clusters. In 1b, the same exemplars are labelled according to the word they are associated with, while colours show their sub-phonemic classes. The acoustic space is three-dimensional (s_1, s_2, s_3) based on FPCA scores (section 3.2.2); left and right views are projections on the (s_1, s_2) and (s_1, s_3) plane, respectively.

There are many other types of phonetic variants that cannot be cleanly defined by the three cases detailed above, partly for the reason, as Scobbie and Stuart-Smith (2008, p. 87) note, that "contrast must be treated as a gradient phenomenon at the phonological level, with membership of a phonemic inventory being a matter of degree". Thus whether or not some of the subtle quasi-phonemic distinctions documented by Scobbie and Stuart-Smith (2008) such as the diphthong differences in *days* vs. *daze* in Northern Irish English or the initial vowel differences in London English between /aʷ/ in *molar* vs. /ɒʷ/ in *roller* fall into different sub-phonemic classes depends entirely on the sensitivity of the acoustic parameterisation to such differences: if the acoustic

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differences are sufficiently large so that these variants are classified following the GMM into different acoustic clusters, then they would also be in different sub-phonemic classes (also because the sub-phonemic classes map to different sets of words).

2.2.3 Memory Management

Revisions were made to the way that exemplars are stored and erased from memory. The changes were made with regard to two main criteria. On the one hand, there has to be a sufficient number of exemplars to make the Gaussians (and Gaussian Mixtures) stable. On the other hand, too many exemplars stored in memory minimise the influence of newly absorbed exemplars and thereby prohibit change. Two measures were implemented so that the agents' memories are filled with an appropriate number of exemplars per word in the beginning and are kept at the same size throughout the simulation. In the case in which there are very few tokens from real speakers available for the initialisation of agents, the number of tokens can be increased by applying SMOTE (Chawla et al., 2002). This is a standard resampling technique that was adapted to the ABM architecture as explained in Appendix D. When the agents start interacting and accepting each others' exemplars, their memories will grow in size. In order to control the memory size, the model allows agent listeners to forget exemplars, i.e. when an agent listener has accepted and memorised a new exemplar, another exemplar of the same word class is removed from their memory. The only constraint on forgetting is that agents cannot forget word classes, i.e., removal is blocked if the deletion of an exemplar would lead to a decrease in the number of exemplars of the word class with respect to the number of exemplars with which the word class was initialised.

3 Simulating a Phonetic Shift

In this section, the revised IP model (section 2) is applied to /u/-fronting in Standard Southern British English (SSBE) that had previously been analysed with the earlier IP model in Harrington and Schiel (2017). The purpose of the study reported in this section is to investigate whether a similar result can be obtained with the modified architecture described in section 2. It was important to check whether established results can be upheld with the revised IP model and especially so if no phonological change was expected, as is the case for /u/-fronting in SSBE.

3.1 Introduction

The study reported by Harrington and Schiel (2017) was concerned with the effects of group contact on phonetic variants of the same phonological category. The general theory to be tested was that if there are two groups of speakers, A and B, that come into contact with each other, and if B's phonetic variants are skewed towards A's in some acoustic space, then when A and B communicate with each other, B's phonetic variants will shift towards A. The theory was tested using an agent-based model in Harrington and Schiel (2017) to simulate contact between two groups of SSBE speakers, one older and one younger, who produced GOOSE vowels with retracted and fronted variants, respectively. This difference between these two groups is a consequence of a sound change in progress in SSBE (Alderton, 2020; Bauer, 1985; Harrington et al., 2008; S. Hawkins & Midgley, 2005; Henton, 1983) which is also in evidence in many other varieties of English (Cox & Palethorpe, 2001; Fridland, 2008; Gordon, 2004; Haddican et al., 2013; Jansen, 2018; Kerswill & Williams, 2005; Koops, 2010) by which variants of GOOSE have been shifting to the front of the vowel space in the last few decades. Older SSBE speakers in Harrington and Schiel (2017) had retracted variants of /u/ in neutral contexts such as *who* (in which there is a minimal coarticulatory influence from the preceding consonant) and in consonantal contexts that induce F2-lowering such as *boo* or *woo*. Older SSBE speakers also had fronted variants following either /j/ (*beauty*) or alveolars (*soon*, *too*). Younger speakers by contrast had fronted variants in all contexts, presumably because for them the /u/-fronting sound change was at an advanced stage (Harrington et al., 2008).

The prediction in Harrington and Schiel (2017) was that, because older speakers' /u/ with a back and a front variant was skewed towards younger speakers' /u/ with only a fronted variant, the shift should be asymmetric such that older speakers' /u/ shifts more towards younger speakers' /u/ than vice versa. This prediction was supported by the outcome of simulated interaction between members of the two groups using the agent-based model in Harrington and Schiel (2017, p. 430). Consistent with the original hypotheses by Harrington and Schiel

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(2017), the more specific predictions regarding the outcome of the interactions between agents representing younger and older SSBE speakers in the agent-based model were that:

1. Older agents /ju, u/ should shift more in the direction of younger agents' /ju, u/ than the other way round.
2. There should be no phonetic change to /i/ (given that it is not skewed in any direction).
3. There should be no phonological changes and no evidence of any splits or mergers.

3.2 Method

3.2.1 Data

The data used in the first experimental part of this paper are the same subset of the data from Harrington et al. (2008) that was used in Harrington and Schiel (2017). For convenience, we provide a brief description of the data here, but more details on the processing procedures can be found in Harrington and Schiel (2017). The data consists of utterances from 22 speakers of Standard Southern British English, equally divided into two age groups (younger: 19-21 years old, older: 56-88 years old). The speakers produced up to ten repetitions of eleven isolated monosyllabic words that contained either /i/ (e.g. *feed*) or /u/ (e.g. *food*) or /ju/ (e.g. *feud*; see Table 1 in Harrington and Schiel (2017) for the full list of words).[3] The total number of productions was 11 (word classes) × 10 (repetitions) × 22 (speakers) = 2420. However, 30 of these were excluded due to mispronunciations, leaving 2390 productions. All /i, u, ju/ vowels were segmented between their acoustic onset and offset without any further sub-segmentation of /ju/. Their first four formant frequencies were calculated using a frame shift of 5 ms and a 30 ms Blackman window. If necessary, formant tracking errors such as the misidentification of F3 for F2 in back vowels were manually corrected. The formant values in Hertz were converted to Bark using the formula in Traunmüller (1990). The formant trajectories were linearly time-normalised to 11 equally spaced data points between the acoustic vowel onset and offset.

3.2.2 Parameterisation

In contrast to Harrington and Schiel (2017) who used the Discrete Cosine Transform (DCT), the time-normalised F1 and F2 trajectories were parameterised using Functional Principal Component Analysis (FPCA, Gubian et al., 2015; Ramsay & Silverman, 2005) in this study. FPCA identifies the main modes of variation in functional, i.e. time-varying, data. The resulting curve parametrisation is usually more compact than the one obtained via DCT for two reasons. First, while DCT coefficients encode global characteristics of curves (i.e., mean, slope, curvature), PC scores are data-specific, i.e. their interpretation changes depending on the input data. Second, FPCA can handle multidimensional data. That is, F1 and F2 trajectories can be represented by e.g. just three Principal Component scores, each modulating *both* F1 and F2 instead of by six DCT coefficients (three each for F1 and F2, respectively).

The following formulae define the FPCA-based parametrisation of the formant trajectories:

$$F1_i(t) \approx \mu_{F1}(t) + \sum_{k=1}^{K} s_{k,i} \cdot PCk_{F1}(t) \tag{1a}$$

$$F2_i(t) \approx \mu_{F2}(t) + \sum_{k=1}^{K} s_{k,i} \cdot PCk_{F2}(t) \tag{1b}$$

$F1_i(t)$ and $F2_i(t)$ are the time-varying F1 and F2 trajectories respectively of the ith vowel production ($i = 1, 2, \ldots, N = 2390$), $\mu_{F1}(t)$ and $\mu_{F2}(t)$ are the mean F1 and F2 trajectories, $PCk_{F1}(t)$ and $PCk_{F2}(t)$ are time-varying Principal Components (PCs, $k = 1, 2, \ldots, K$), and $s_{k,i}$ are PC scores that modulate PCk and are different for each of the $N = 2390$ vowel productions. That is, the F1 and F2 trajectories are linearly decomposed into a set of PCs added to the mean trajectories. The PC scores are the same in 1a and 1b because each PC score modulates both F1 *and* F2 to capture possible systematic co-variations. The scores for the first $K = 3$ PCs were used as acoustic exemplars in the simulation, i.e. the agents' acoustic space was 3-dimensional.

[3] The vowels /i, ju, u/ were all long vowels. We refrain from using the length mark for convenience.

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3.2.3 Model Settings

The architecture of the agent-based model and communication between agents was as described in section 2. There were 22 agents, one for each of the 11 younger and 11 older participants described in section 3.2.1. Each agent was initialised with the word classes and acoustic exemplars of the corresponding speaker. Prior to any interaction, each agent computed the sub-phonological classes using GMM and NMF in the manner described in section 2.2.2. The agents communicated across age groups (a younger with an older agent or vice versa) in a total of 500,000 interactions. This procedure was repeated 10 times (i.e. 10 runs) in order to test for the robustness of the results. Both the absolute (Mahalanobis distance) and relative (maximum posterior probability) memorisation criteria were applied to determine whether an agent accepted a newly perceived exemplar. The threshold for the Mahalanobis distance was set to a value corresponding to a centred region of the multi-dimensional Gaussian distribution (ellipsoid) containing 95% of its total probability mass.[4] Lastly, the algorithms for acoustic clustering and sub-phonemic grouping were used throughout the simulation even though no sub-phonemic restructuring was expected. When an agent had accepted 100 new exemplars, it re-estimated the GMMs and re-applied NMF. A sub-phonemic group had to be at least 75% pure (see section 2.2.1) for it to be identified as its own sub-phoneme, which means that three quarters of all exemplars in a sub-phoneme had to be associated with the same distinct subset of word classes.

3.2.4 Quantification of Change

The acoustic and sub-phonemic levels were analysed and interpreted separately. For each of them it was determined whether there had been any change in the *post-run* relative to the *baseline*. The baseline was the state of the agents after initialisation and before any interaction. The post-run was the state after the given number of interactions had taken place.

The analysis of the acoustic level was conducted in the domain of time-varying F1 and F2 trajectories. These were obtained by applying FPCA-based reconstruction formulae (1a) and (1b) to the memorised exemplars (consisting of the three PC scores s_1, s_2, and s_3). For quantifying acoustic changes, generalised additive mixed models (GAMMs) were applied separately to the reconstructed F1 and F2. The fixed (smooth) factors were age group (younger, older), state (baseline, post-run), and canonical phoneme (/i, u, ju/). The random effects were agent (22 levels), word (11 levels), and run (10 levels). The full specification of GAMM equations, detailed hypotheses and statistical results can be found in Appendix E.

The sub-phonemic level was analysed using a number of indirect measurements. First, sub-phonemic similarity matrices S were computed. Given the agent memories at a certain point in time and given the set of word classes $\{w_i\}, i = 1, \ldots, N_w$, S is a $N_w \times N_w$ symmetric matrix where $S[i, j]$ is the proportion of agents who grouped words w_i and w_j together in the same sub-phoneme (this shall be shown in more detail in Figure 3). In simple terms, S provides a picture of the population-wise subdivision of the vocabulary into sub-phonemic classes: the closer its values are to 0 or 1 the more the agent-specific sub-phonemic classes agree with one another; vice versa, the closer to 0.5 the more diverse the sub-phonemic classes are across the population. S also provided a way to monitor the degree of consistency across separate independent simulation runs, which generally starts very high and decreases over time. Consistency across separate runs was quantified by computing the correlations between pairs of S matrices from different runs at the same point in simulation time. These correlations were computed using the Mantel test (Mantel, 1967), which takes into account the statistical dependencies induced by the structure of a similarity matrix.

Changes to phonological categorisation were determined by calculating the degree to which sub-phonemic classes matched canonical phonemes. This was done by calculating *canonical agreement* A, which we define as:

$$A = \frac{1}{N_w} \cdot \sum_p N_{\text{majority}}(p), \tag{2}$$

where N_w is the total number of word classes and $N_{\text{majority}}(p)$ is the number of unique word classes in the sub-phoneme p that belong to the canonical phoneme, which counts the largest number of unique word classes in

[4] For a 3-dimensional Gaussian the actual Mahalanobis distance corresponds to the square root of the quantile $q_{.95}$ of a χ^2_3 distribution, ≈ 2.80.

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p.[5] For instance, the canonical agreement for the system of three sub-phonemes shown in Figure 1b is 1, since they match exactly with the three canonical phonemes /i, ju, u/. Suppose instead that sub-phoneme p_1 contains all 4 /i/ words, p_2 all 3 /ju/ words and also the /u/ word *soup* and p_3 the remaining 3 /u/ words (11 word classes in total). Then the sum in Eq. (2) equals 4 + 3 + 3, hence $A = \frac{10}{11} \approx 0.9$.

Along with agreement, the number of sub-phonemic classes N_p was computed too. This is because in limit cases agreement alone may convey the wrong picture. For instance, a perfect agreement of 1 can be trivially obtained by forming one sub-phonemic class per word class. The question to be considered was whether agreement and the number of sub-phonemic classes differed between baseline and post-run, and whether patterns differed between the two speaker groups. The changes in agreement were tested with the following binomial regression mixed model:

$$\sum_p N_{\text{majority}}(p)_i \sim \mathit{Binomial}(\theta_i, N_w)$$
$$\text{logit}(\theta_i) = \beta_0 + \beta_{\texttt{gr}} \cdot \texttt{gr}_i + \beta_{\texttt{st}} \cdot \texttt{st}_i + \beta_{\texttt{gr:st}} \cdot \texttt{gr}_i \cdot \texttt{st}_i + u_{\texttt{ag}[i],1} + u_{\texttt{ag}[i],2} \cdot \texttt{st}_i, \tag{3}$$

where i is the running index on agents × states, `gr` and `st` are binary variables encoding group (older = 0, younger = 1) and state (baseline = 0, post-run = 1), $\texttt{ag}[i]$ identifies the agent corresponding to index i, and u is a two-dimensional Gaussian variable with zero mean which models a random intercept for agent (22 levels) and a random slope by state (plus their correlation). [6] Model (3) was applied on one randomly chosen simulation run out of the 10 computed, as opposed to using all of them and introducing run as further random factor, since the correlation between sub-phonemic similarity matrices S was always very high (> 0.95).[7]

Similarly, changes in the number of sub-phonemic classes were tested. In this case the number of sub-phonemes minus one was modelled by a Poisson distribution[8] whose parameter depends on group and state:

$$N_{p,i} - 1 \sim \mathit{Poisson}(\lambda_i)$$
$$\log(\lambda_i) = \beta_0 + \beta_{\texttt{gr}} \cdot \texttt{gr}_i + \beta_{\texttt{st}} \cdot \texttt{st}_i + \beta_{\texttt{gr:st}} \cdot \texttt{gr}_i \cdot \texttt{st}_i, \tag{4}$$

where i is the running index on agents × states.

3.3 Results

The estimated mean F2 trajectories in Figure 2, which are more relevant to the sound change in question than F1, show a more fronted /u/ and a slightly more fronted /ju/ for older agents post-run (green) relative to the baseline (black). The same figure shows very little change for younger agents except for a slightly lower F2 after the simulation compared to before in both /u/ and /ju/. The results of the statistical analysis with GAMMs confirmed a significant shift of older agents' /u/ in the direction of younger agents' /u/ in F2 ($edf = 8.9$, $rdf = 8.9$, $F = 41.7$, $p < 0.001$).[9] Contrary to the predictions, the GAMMs showed a trend towards a significant retraction of younger agents' /u/ ($edf = 5.3$, $rdf = 5.4$, $F = 2.3$, $p = 0.05$). A test on the difference between these two changes confirmed that the fronting of older agents' /u/ was significantly larger than the retraction of younger agents' /u/ ($edf = 8.5$, $rdf = 8.6$, $F = 10.7$, $p < 0.001$). The shift from older agents' /ju/ towards that of younger agents was not significant, but younger agents' /ju/ shifted significantly towards that of older agents ($edf = 5.1$, $rdf = 5.2$, $F = 2.5$, $p < 0.05$). However, a test on the difference between these two changes showed no significance, i.e. there is insufficient evidence to show that the retraction of younger agents' /ju/ was greater than the fronting of older

[5] Using the notation introduced in Appendix C, $N_{\text{majority}}(p) = \max_q (|\{w : p(w) = p, q(w) = q\}|)$, where q indicates canonical phoneme. The definition in (2) corresponds to purity of clustering with respect to known categories, e.g. see the `purity` function in R package *NMF* (Gaujoux & Seoighe, 2010). We used the term agreement to avoid confusion with purity as defined in section 2.2.1 and appendix C.2.

[6] Using R package *lme4* (Bates et al., 2015), Eq. (3) is specified as: `glmer(c(N_majority, N_words - N_majority) ~ gr * st + (st | ag), family = binomial, link = logit)`.

[7] Using all runs and introducing run as further random factor in model (3) has only negative effects on convergence of estimation, precisely because the inter-run variance is too small.

[8] The Poisson distribution models counts from zero to infinity, while the number of sub-phonemes N_p can vary between one and the number of word classes N_w. Subtracting one from N_p matches the minimum value to the Poisson distribution, while the upper limit is high enough ($N_w - 1 = 10$ in the SSBE case) to be ignored in the model.

[9] *edf* stands for the estimated degrees of freedom, *rdf* for the reference degrees of freedom.

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agents' /ju/. Compatibly with the predictions, the GAMMs showed no significant change to the F2 trajectory of /i/ for either age group.

The GAMMs computed for F1 (see Appendix E.1 for visualisation) showed a significant raising of the trajectory for older agents' /u/ ($edf = 2.0$, $rdf = 2.0$, $F = 5.4$, $p < 0.01$), which is a change in the direction of younger agents' F1 trajectory for /u/, and no other significant change. In any case, the changes in F1 were always much smaller than those in F2.

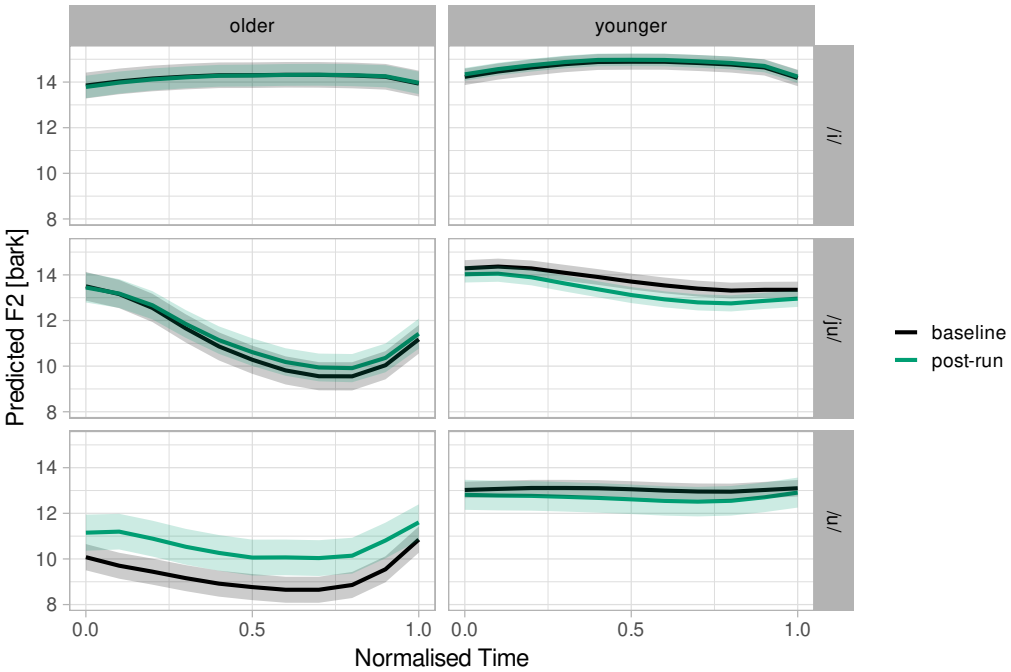

Figure 2: Predicted F2 trajectories of /i/, /ju/, and /u/, both before (black) and after the simulation (green), with confidence bands.

Figure 3 shows the association between the word and sub-phonemic classes, separately by age group (columns) and by state (before/after the simulation, rows), for one out of the ten runs. The darker a coloured tile, the higher the proportion of agents who grouped a given pair of words together in the same sub-phoneme. The three diagonal, dark purple squares in each of the four panels indicate that the association between exemplars and sub-phonemic classes was predictably related to canonical phonemes in both the baseline and post-run and for both age groups. That is, words containing /i/ were grouped almost exclusively with other /i/-words, and the same holds for /u/-words and /ju/-words. The main exception to this pattern was the word *soup* which was sometimes grouped with /ju/- instead of with /u/-words by older agents before the simulation, and with both /u/- and /ju/-words after the simulation. This undoubtedly came about because the high F2 locus of alveolars (Cooper et al., 1952; Iskarous et al., 2010) causes a strong acoustic resemblance to the high F2 onset and falling onset transition in /ju/-words.

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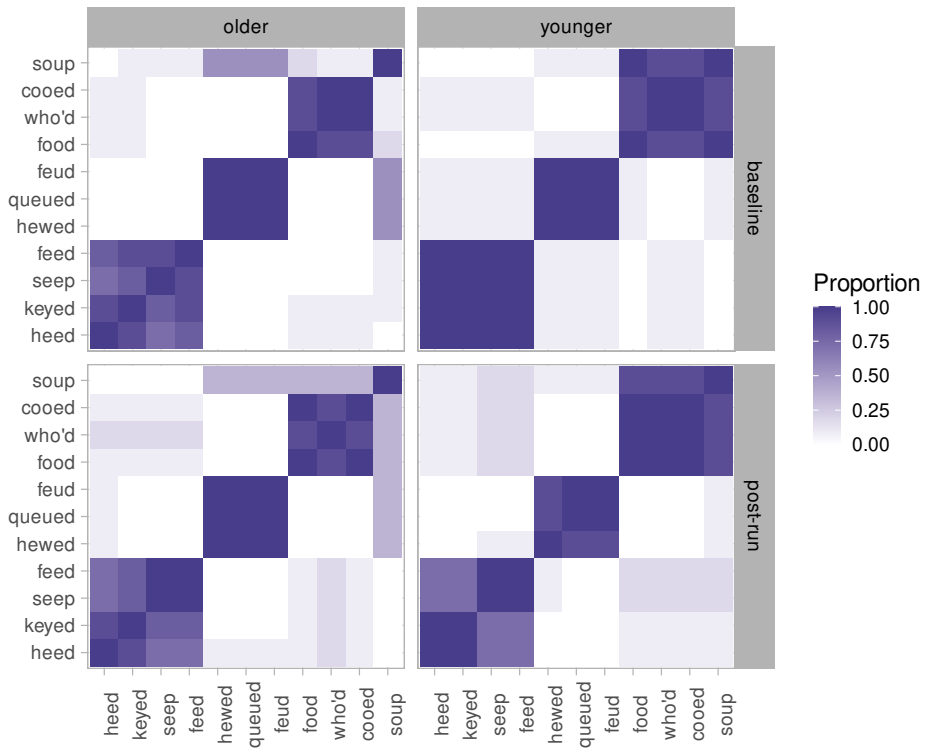

Figure 3: Proportion of agents (colour-coded) who grouped together two word types in one sub-phoneme. The words on the axes are ordered by canonical phonemes (first /i/, then /ju/, and lastly /u/). This is the result for a single run of this simulation.

The Poisson model in Eq. (4) and the corresponding post-hoc tests predicted 3.5 and 3.4 sub-phonemes for older, and 2.8 and 3.1 sub-phonemes for younger agents in the baseline and post-run, respectively. This shows that the number of sub-phonemes was close to the actual number of canonical phonemes in both age groups and both before and after the simulation, even though the agents were never given any information about the canonical phonemes. Neither age group nor simulation state had a significant effect on the number of sub-phonemes.

The expected canonical agreement (Eq. (2)) predicted by the generalised mixed model in Eq. (3) was 0.95 and 0.94 for older, and 0.98 and 0.96 for younger agents before and after the simulation, respectively. These results and subsequent post-hoc tests reveal the following three main findings. Firstly, and consistently with hypothesis 3, agreement was unaffected by state: that is, the degree to which the mapping between exemplars and sub-phonemic classes was related to canonical phonemes was no different in the baseline compared to post-run. Secondly, there were no significant differences between younger and older agents' canonical agreement. Thirdly, and perhaps most importantly, the predicted value of agreement was very high (≥ 0.94) in all cases. Thus overall, although the mapping from exemplars to sub-phonemic classes was completely unsupervised, it was nevertheless very strongly in agreement with the division of exemplars into canonical phonemes.

3.4 Discussion

The purpose of the analysis in this section was to compare the agent-based model described in section 2 with that of Harrington and Schiel (2017) when applied to the same acoustic data. The application of the revised model in section 2 to this dataset differed from that of Harrington and Schiel (2017) in three major ways: phonological classes were derived in a data-driven way rather than being fixed; the acoustic parameters were based on time-varying F1 and F2 together rather than on just F2; and there was a different strategy for deleting exemplars from memory.

The results for the acoustic changes due to this agent-based model are broadly in agreement with those of Harrington and Schiel (2017). In particular, in both models there was an asymmetric shift such that older agents' /u/ shifted more towards that of younger agents than vice versa. Both models were also in agreement in showing

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no change to /i/ in either older or younger agents. The main difference between the outcomes of agent interaction in the two models was in /ju/ which, in contrast to the model in 2, showed greater F2-raising for older agents and no significant F2-lowering for younger agents in Harrington and Schiel (2017). Visually, however, there was some F2-lowering in younger agents' /ju/, as can be seen from Fig. 8 in Harrington and Schiel (2017). Both in Harrington and Schiel (2017) and in the model tested here, this unexpected change in younger agents' /ju/ most likely comes about because of a repulsion effect between /i/ and /ju/ which are very similar acoustically. Hence, the relative memorisation criterion prevents acceptance of exemplars in the ambiguous acoustic space between /i/ and /ju/ and allows only the memorisation of /ju/-exemplars with lower F2.

In Harrington and Schiel (2017) the phonological classes were pre-assigned and immutable canonical (classical) phonemes (/i, ju, u/). The phonological classes in the present model were by contrast derived in an entirely data-driven way: above all, the derived sub-phonemic classes could split and merge during interaction (similarly to Stevens et al., 2019). Despite these very different phonological architectures, there was close agreement between the two models: in particular, in this revised model the information contained in the derived sub-phonemic classes corresponded almost entirely with the division of the exemplar space into classical phonemes.

These two null results – the lack of acoustic change in /i/ and the consistency in phonological classification from initialisation to the end of the run – show that neither phonetic nor phonological change is an inevitable consequence of the agent-based model tested here. An agent-based model that always resulted in change would not be compatible with the evidence showing that it is only a fraction of phonetic variation that actually becomes sound change (Baker et al., 2011; Ohala, 1993; Sóskuthy, 2015).

4 Simulating a Merger

This study was concerned with testing whether the same ABM architecture used to simulate the phonetic shift of SSBE speakers towards a fronted GOOSE vowel in section 3 could be employed to predict the merger of the lexical sets NEAR and SQUARE in New Zealand English. In contrast to /u/-fronting in SSBE, the diphthong merger in NZE has both acoustic and phonological ramifications both of which should be predicted by the revised IP model given that it provides the possibility of re-organising the agents' phonological level by means of GMM and NMF (section 2.2.1).

4.1 Introduction

The merger of /ɪə/ and /eə/ in NZE was first described roughly half a century ago, based largely on auditory impressions (P. Hawkins, 1973; Turner, 1966, 1970; Wells, 1982). Later studies found that the merger had begun long before the 1970s, possibly even in the 19th century (Gordon & Maclagan, 1985; Gordon & Trudgill, 1999). Today, the /ɪə, eə/ merger is considered a characteristic feature of NZE in contrast with RP (Gordon, 2004) and Australian English (Holmes, 1995; Watson et al., 1998), and has been the subject of many quantitative acoustic analyses in the last decades.

In their seminal research project called "The Canterbury NEAR / SQUARE study", Gordon and Maclagan (1985, 1990, 2001) and Maclagan and Gordon (1996) recorded groups of 14-year old NZE speakers in 1983, 1988, 1993, and 1998. Over this time span there was a significant increase in the teenagers' usage of a merged diphthong for both NEAR and SQUARE. The spread of the merger amongst the teenagers followed a sigmoidal shape: Only 16% of all participants in 1983 and 22% in 1988 produced the merger consistently, but in the five years between 1988 and 1993 the merger suddenly took hold in the speech of 76% of all recorded teenagers. The data from 1998 showed a consolidation of this trend, i.e. 80% of the participants produced a merged variant. Gordon and Maclagan (2001) were also able to observe a change in the direction of the merger in their longitudinal data: While 32% of teenagers produced a diphthong closer to /eə/ and 19% produced a diphthong closer to /ɪə/ in 1983,[10] the reverse was the case in 1988, and by 1998, only 1% of participants produced /eə/ for both NEAR and SQUARE, while 71% produced /ɪə/.

The NZE centring diphthongs were characterised by a high degree of acoustic variability for a long time before the merger was documented for the whole population (e.g., consider the contradictory accounts by Trudgill and

[10] The remaining 49% of teenagers produced a distinction between NEAR and SQUARE.

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Hannah, 1994; Turner, 1970; Watson et al., 2000). There is now abundant evidence showing that the first element of /eə/ was gradually raised such that most contemporary NZE speakers realise both 'hear' and 'hair' as /hɪə/ (Gordon & Maclagan, 2000; Hay et al., 2008; Hay et al., 2006; Kennedy, 2004; Maclagan & Hay, 2007; Rae & Warren, 2002; Warren & Hay, 2006; Watson et al., 1998). An acoustic analysis of a larger set of the same data to be used in the present study showed that younger NZE speakers made hardly any distinction between /ɪə/ and /eə/ (Gubian et al., 2019). Older speakers, however, still produced a slightly more peripheral /eə/ (i.e., with lower F1 and higher F2), thereby maintaining some acoustic separation between the diphthongs. Given the apparent-time approach, the results from that study provided support for the direction of change from /eə/ towards /ɪə/. Gubian et al. (2019) also found that the merger was accompanied by an increase in formant tilt, i.e. a change from a more monophthongal /eə/ towards a more diphthongal /ɪə/ (also see Maclagan and Gordon, 1996).

The study presented in this section aims to model the diphthong merger by means of interactions between two agent groups. Similarly to the study in section 3, younger NZE speakers represent a more advanced or even completed stage of the sound change whereas older speakers represent an older state of NZE in which a contrast between /ɪə/ and /eə/ still existed. It was predicted that the interactions between younger and older NZE speakers should result in an acoustic and phonological merger of /ɪə/ and /eə/. The more specific hypotheses to be tested were that:

1. Older agents' /eə/ should shift more in the direction of younger agents' /eə/ than the other way round.
2. There should be no phonetic change to /ɪə/.
3. The sub-phonemes should agree less with the canonical separation of /ɪə/ and /eə/ after than before the simulation, especially amongst older agents.

4.2 Method

4.2.1 Data & Parameterisation

The material and speakers are a subset of the data used in Gubian et al. (2019) which was taken from the New Zealand Spoken English Database recorded in 1999/2000. There were 48 speakers, equally divided into two age groups (older: 46-60 years, younger: 18-30 years), two regional groups (Hamilton, Wellington) as well as two genders (male, female). These speakers had produced 53 words once or twice in isolation or short sentences. There were 30 /eə/ and 23 /ɪə/ word classes. With the exception of *questionnaire*, *fun-fair*, and *thoroughfare*, the diphthongs occurred in the syllable with primary lexical stress. There were 1245 productions of /ɪə/ and 1530 productions of /eə/, resulting in a total of 2775 diphthongs to be analysed.

The words were automatically segmented using MAUS (Kisler et al., 2017) and the first five formant frequencies were computed using a frame shift of 6.25 ms and a Gaussian window of 25 ms. The segment boundaries and formant frequencies were manually corrected if necessary. The F1 and F2 trajectories of /ɪə/ and /eə/ were z-score normalised with respect to the speaker's /i, ɔ, ɐ/ and linearly time-normalised to 11 time points. Subsequently, the formant trajectories were parameterised by means of FPCA, as described in section 3.2.2. The interpretation of the first $K = 3$ Principal Components for this New Zealand English dataset is described in Gubian et al. (2019).

4.2.2 Model Settings

The IP model was applied to the NZE data using the same settings as were used to simulate the phonetic shift in section 3. That is, there were 48 agents, 24 of them representing the younger and 24 representing the older speakers. After initialising each agent with the word classes and exemplars of one of the speakers, SMOTE was applied to increase the number of tokens per word (see Appendix D for more details). This was necessary because there were a maximum of two tokens per word per speaker (agent) which renders the computation of Gaussians and Gaussian Mixtures too unstable. The application of SMOTE to each agent resulted in a tenfold increase in the number of tokens per word. The agents then computed the phonological classes using GMM and NMF before starting to communicate with agents of the opposite age group in a total of 300,000 interactions. The simulation was repeated 10 times. Everything else was exactly as detailed in 3.2.3. Any acoustic and phonological changes were quantified in the same way as in 3.2.4.

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4.3 Results

The predicted F1 trajectories in Figure 4 show that older agents had a more peripheral /eə/ post-run (green) compared to the baseline (black) as shown by the substantially lowered F1 trajectory. There were hardly any visual changes in younger agents' diphthongs and to older agents' /ɪə/ apart from a slight reduction in tilt. In line with hypothesis 1, the GAMMs confirmed a significant change in older agents' /eə/ ($edf = 5.0$, $rdf = 5.3$, $F = 4.0$, $p < 0.01$), while the change in younger agents' /eə/ was not significant ($edf = 2.0$, $rdf = 2.0$, $F = 2.8$, $p = 0.06$). However, a test of the difference between those two changes did not reach significance, although the predicted trend goes in the expected direction: that is, the change in older speakers was greater than in younger speakers. The GAMMs also showed that the age groups differed significantly in their pronunciation of /eə/ before the simulation ($edf = 5.8$, $rdf = 6.0$, $F = 6.4$, $p < 0.001$), but were not statistically different after the simulation for either diphthong. Contrary to the predictions, the change in older agents' /ɪə/ was also significant ($edf = 2.5$, $rdf = 2.6$, $F = 5.1$, $p < 0.01$).

The GAMMs computed for F2 (see Appendix E.2 for visualisation) showed that there was a significant difference between the two age groups' /eə/ before ($edf = 4.6$, $rdf = 4.9$, $F = 5.7$, $p < 0.001$), but not after the simulation. Furthermore there were some minor, yet significant changes in F2 tilt for younger agents' /ɪə/ ($edf = 2.1$, $rdf = 2.1$, $F = 4.6$, $p < 0.05$) and /eə/ ($edf = 2.0$, $rdf = 2.0$, $F = 3.9$, $p < 0.05$) as well as older agents' /ɪə/ ($edf = 2.0$, $rdf = 2.0$, $F = 5.9$, $p < 0.01$).

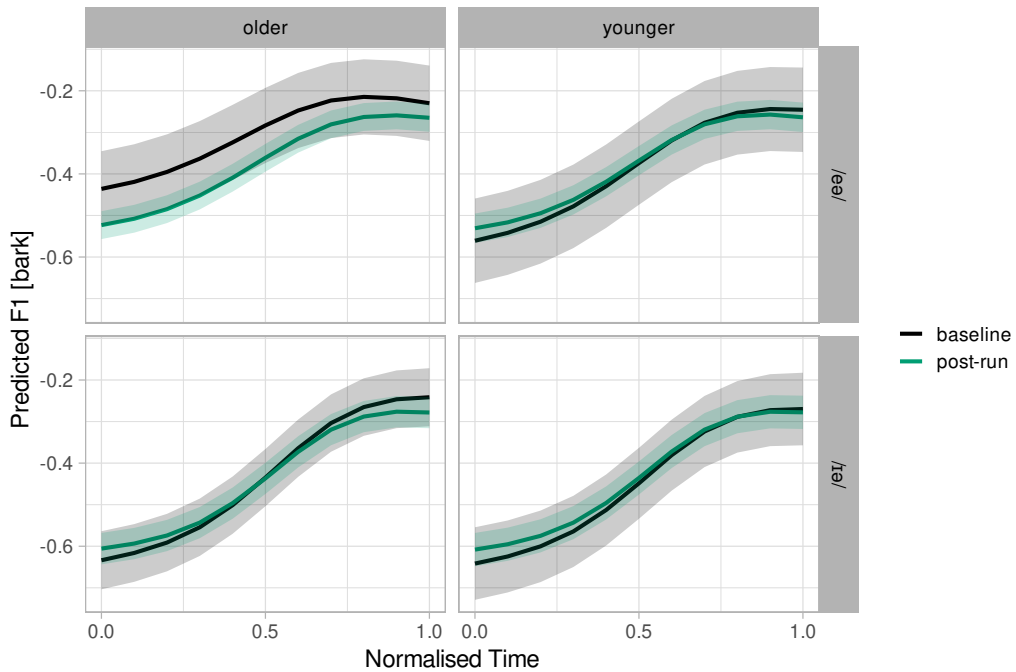

Figure 4: Predicted F1 trajectories of /ɪə/ and /eə/, both before (black) and after the simulation (green), with confidence bands.

For one out of the ten simulation runs, Figure 5 shows the proportion of agents who grouped pairs of words into a sub-phoneme. The 53 words on either axis are not shown and are instead grouped by canonical phonemes. If agents grouped words together in sub-phonemes according to the canonical distinction between /ɪə/ and /eə/, the panels would show two large dark purple squares, one on the bottom left and the other on the top right. Before the agents started communicating with each other (top row), younger agents did not show any discernible pattern of grouping words according to the canonical (classical) phonemes /ɪə/ and /eə/. Older agents, on the other hand, showed a mild trend towards grouping words according to classical phonemes, i.e. the squares on the bottom left and top right are slightly darker overall than the squares in the top left and bottom right. In the post-run there was a clear grouping pattern which was the same in both age groups and indicates that agents formed mainly two sub-phonemes which bore no relationship to the separation into canonical phonemes.

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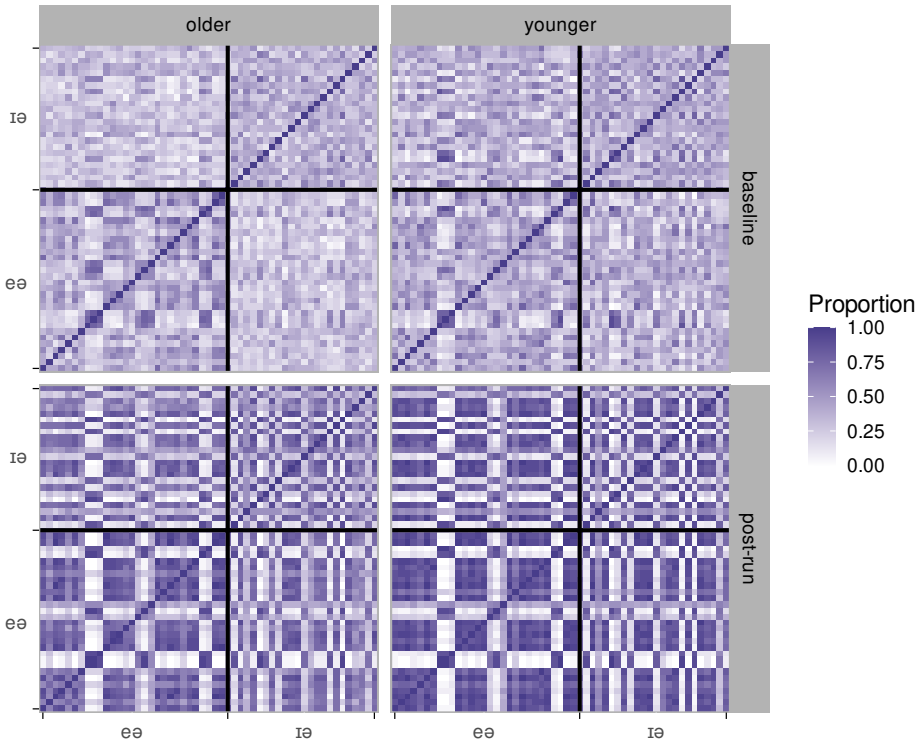

Figure 5: Proportion of agents (colour-coded) who grouped together two word types in one sub-phoneme for one out of ten runs of this simulation. The 53 word types on the axes are ordered by canonical phonemes as indicated (the word labels are not shown for reasons of clarity).

The generalised mixed model in Eq. (3) and subsequent post-hoc tests with `emmeans` showed that the predicted agreement between sub-phonemes and canonical phonemes was 0.68 for older agents and 0.61 for younger agents before the simulation, i.e. older agents adhered more to the canonical phonological separation than younger agents. In the post-run, the predicted agreement was 0.61 for older and 0.58 for younger agents, showing that the canonical agreement decreased in both age groups. The canonical agreement was overall much lower here than in the SSBE simulation and younger NZE agents reached almost chance level in the post-run.[11] Importantly, this means that the agents incrementally generated sub-phonemes that contained a mixture of /eə/- and /ɪə/-words. In line with hypothesis 3, the contrast in canonical agreement between older and younger agents was significant before ($z = 3.0$, $p < 0.01$), but not after the simulation. There was also a significant contrast between the baseline and post-run for older ($z = 3.3$, $p < 0.001$), but not for younger agents.

The Poisson model in Eq. (4) and corresponding post-hoc tests predicted 3.3 and 2.3 sub-phonemes for older, and 2.8 and 2.2 sub-phonemes for younger agents before and after the simulation, respectively. That is, the number of sub-phonemes decreased in both age groups over the course of the interactions from roughly three to two sub-phonemes. The difference in number of sub-phonemes between the baseline and post-run was significant for older ($z = 2.7$, $p < 0.01$), but not for younger agents. In combination with the decrease in canonical agreement, these findings indicate a phonological merger of the two diphthongs under investigation.

A closer investigation of the post-run word grouping into sub-phonemes (Figure 5, bottom row) showed that the majority of agents formed two sub-phonemes which coherently consisted of exemplars from 48 out of 53 word classes. Table 1 shows that one sub-phoneme contained words in which the diphthong predominantly appeared before /d/, /t/, /z/, or /s/, and the other contained words in which the diphthong appeared in word- or syllable-final position. In order to quantify the emergence of this grouping pattern, formula 2 was adapted to compute the agreement between sub-phonemic classes and the two groups in Table 1 and the generalised linear mixed model from Eq. (3) was re-computed to predict the effect of age group and simulation state on the adapted

[11] Since 30 out of 53 word types in the dataset contained /eə/, the chance level, where a sub-phoneme contained a balanced mixture of /eə/- and /ɪə/-words, is $30/53 \approx 0.57$.

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Group 1: before alveolars	veers	/ɪəz/	shared	/eəd/
	fierce	/ɪəs/	compared	/eəd/
	scarcely	/eəs/	here to	/ɪət/
	cares	/eəz/	bear to	/eət/
	cheered	/ɪəd/	dare to	/eət/
	beard	/ɪəd/	sincere condolences*	/ɪək/
	appeared	/ɪəd/	prepare*	/eə/
	chaired	/eəd/	merely *	/ɪəl/
Group 2: in final position	dear old	/ɪəɹ/	experiences	/ɪəɹ/
	sheer fluke	/ɪəf/	nearer	/ɪəɹ/
	gear half	/ɪəh/	staring	/eəɹ/
	Lear was	/ɪəw/	Hungarian	/eəɹ/
	sphere	/ɪə/	Mary	/eəɹ/
	fear	/ɪə/	hilarious	/eəɹ/
	year at, year I	/ɪəɹ/	airing	/eəɹ/
	jeer at	/ɪəɹ/	tearing	/eəɹ/
	weir	/ɪə/	varies	/eəɹ/
	fun-fair is	/eəɹ/	parents	/eəɹ/
	questionnaire about	/eəɹ/	area	/eəɹ/
	rare giant	/eəʤ/	various	/eəɹ/
	wear black	/eəb/	fairly	/eəl/
	aware that	/eəð/	heirloom	/eəl/
	declare	/eə/	years*	/ɪəz/
	mysterious	/ɪəɹ/	arrears*	/ɪəz/

Table 1: Word groups discernible from the bottom row of Figure 5. Five out of the 53 target words were not grouped consistently enough by all agents to appear in this table. Where necessary, the word following the target word is shown as well. For each target word, the diphthong and following consonant were phonemically transcribed. Asterisks indicate words that are in the wrong category according to their transcription.

agreement measure. As could be expected from Figure 5, the model showed a significant effect of state ($z = 6.3$, $p < 0.001$), but not of age group on the adapted agreement. The predicted agreement between sub-phonemes and the context-dependent word groups was 0.78 for older and younger agents before the simulation, and 0.91 for older and 0.93 for younger agents in the post-run.[12] That is, the influence of the diphthongs' phonetic context (i.e., before alveolars or in final position) on how agents grouped word classes into sub-phonemes, while present from the beginning, increased significantly over the course of the simulation.

4.4 Discussion

The aim of the simulation presented in this section was to test whether the revised architecture of the IP model was capable of predicting the NZE NEAR / SQUARE merger using the same settings that were shown to successfully model /u/-fronting in SSBE. The results showed an asymmetric acoustic change in older agents' /eə/ towards that of younger agents. That is, the acoustic merger of the two diphthongs, which was close to completion for younger agents in the baseline, also took hold in the speech of older agents after they had conversed with younger ones. The statistical analysis of the acoustic change provided only a partial confirmation of the expected asymmetry. This may be because the predicted differences and changes were at most around 0.1 barks in size, which is an order of magnitude smaller than those predicted for the SSBE case. This means that differences between younger and older agents' acoustic exemplars were already quite small from the start, which makes it hard for clear patterns to emerge from noise (and be detected statistically). Nevertheless, the older agents' acoustic change from /eə/ towards /ɪə/ provides support for the hypothesis put forward by others (Gordon & Maclagan, 2001; Hay et al., 2006; Rae & Warren, 2002) that the NZE NEAR / SQUARE merger is a merger by approximation.

Furthermore, the regular and unsupervised updates of the agent-specific association between exemplars and sub-phonemes resulted in an increased neutralisation of the phonological contrast between the diphthongs over

[12] Since 32 out of 48 word classes were associated with group 2 in Table 1, the chance level of agreement between these groups and sub-phonemes is $32/48 \approx 0.67$.

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simulation time. This was most apparent again in older agents who, in the beginning, had sub-phonemes that somewhat mirrored the canonical separation of /ɪə/ and /eə/. In addition to the relatively small acoustic changes summarised above, the low canonical agreement for NZE as compared to SSBE agents at the baseline indicates that the merger was quite advanced for most of the NZE agent population (older as well as younger speakers). As shown by e.g. Stanford and Kenny (2013), agent-based models can be sensitive to imbalances in the input data, such as unequal numbers of agents per agent group or disproportional amounts of input from single agents. However, further simulations provided in Appendix F show that the results on NZE presented in section 4.3 were not a mere matter of imbalanced data.

A further (and unexpected) finding from the NZE simulations was the emergence of new phonological classes as a consequence of interaction. More specifically, as the interactions progressed, there was both an increase in the degree of the phonological /eə, ɪə/ merger and in the agreement across agents in their grouping of the centring diphthongs into two new phonological categories. These phonological categories were context-dependent, with one comprising mostly words in which the diphthong was followed by an alveolar (*beard*, *cares*, *merely*) and the other including primarily pre-pausal or pre-/r/ (*caring*) contexts. The statistical analysis showed that this context-dependent grouping of exemplars into sub-phonemes was already evident at the baseline, but became more apparent towards the end of the simulation. The likely reason why these new sub-phonemic classes emerged was because of a coarticulatory effect that was consistent across agents (Gubian et al., 2019). More specifically, when the centring diphthongs occur before an alveolar, the F2 frequency that is already high because of the first /ɪ, e/ component is likely to remain high due to the high F2-locus at just under 2 kHz of alveolars (Fant, 1973; Kewley-Port, 1982; Liberman et al., 1954). By contrast for pre-/r/ and pre-pausal diphthongs, F2 and especially F3 are likely to be lower. Because there is no rapid transition from a high vowel to a following consonantal constriction in pre-pausal position, the vowel is longer and has more time to attain a phonetically central position which are associated with a lower F2 and F3 than preceding alveolars. Because this is a consistent coarticulatory effect across agents, two sub-phonemic classes emerged for the population that map to these two sets of words (also see section 5). Additional simulations in Appendix F showed that the presence of the coarticulatory effect in the NZE data and its subsequent phonologisation was not causally related to the merger of /ɪə/ and /eə/.

5 General Discussion

The aim of this paper was to present a revised version of the IP model and test it on two different sound changes. The first of these was /u/-fronting which is a phonetic change that does not affect the phonological system of the language; the second was a merger between two centring diphthongs requiring a flexible association between acoustic exemplars and phonological classes. As explained in section 2.2.1, the phonology module implemented in the revised IP model consisted of two fully unsupervised, general-purpose learning algorithms and was shown to be able to model both stability and change at the phonological level (Sóskuthy, 2015).

The application of the ABM to /u/-fronting showed a similar pattern of results to that in Harrington and Schiel (2017), i.e., an asymmetric change such that the retracted /u/ from the older speakers shifted more towards the front of the younger speakers' vowel spaces than vice versa. Although the derivation of phonological classes in the revised IP model was entirely bottom-up and data-driven, the simulations with /u/-fronting showed no changes to phonological categorisation. This finding demonstrates, therefore, that a split, merge, or other type of phonological re-organisation is not an inevitable outcome following interaction in the revised IP model.

When the ABM was applied to the New Zealand English data, interaction increased the degree of merger between the centring diphthongs. The merger consisted of the reduction of acoustic differences between /ɪə/ and /eə/ along with the neutralisation of the phonological distinction between the two. This increase in merger during the simulation was to a certain extent caused by a high degree of inconsistency across agents in the way that words were distributed in an acoustic space. An example of a consistent relationship is in the data from /u/-fronting. The consistency across agents in this case is that /i/-words (*heed*, *feed*) and /u/-words (*who*, *food*) were positioned acoustically in phonetically front vs. central or back parts of the vowel space for all agents. For /ɪə, eə/ by contrast, there was no such consistent relationship prior to interaction. Thus, although /ɪə, eə/ words differed to a certain extent in phonetic height for some agents (although always with some overlap between them), for other agents this was not necessarily the case. Consequently, the consistency differed across agents with which words were

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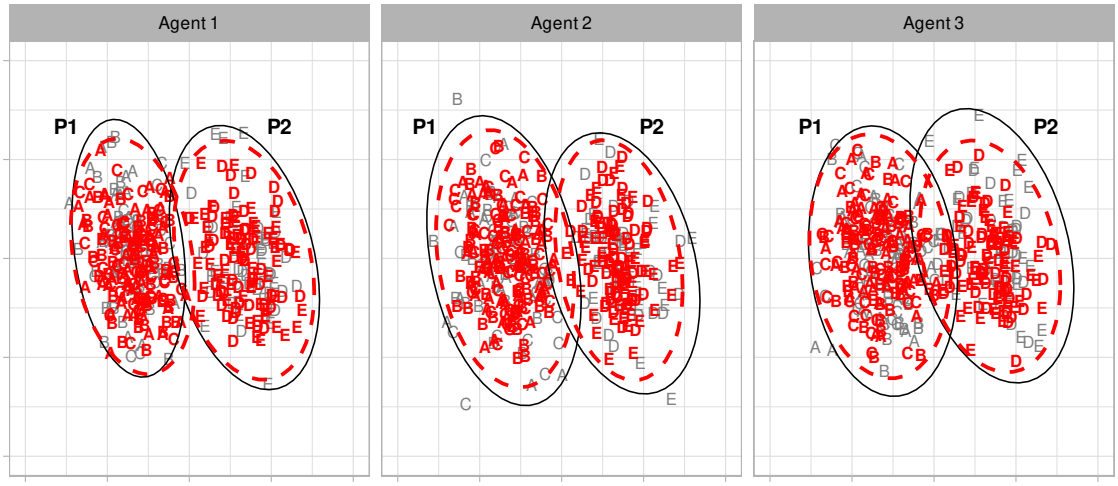
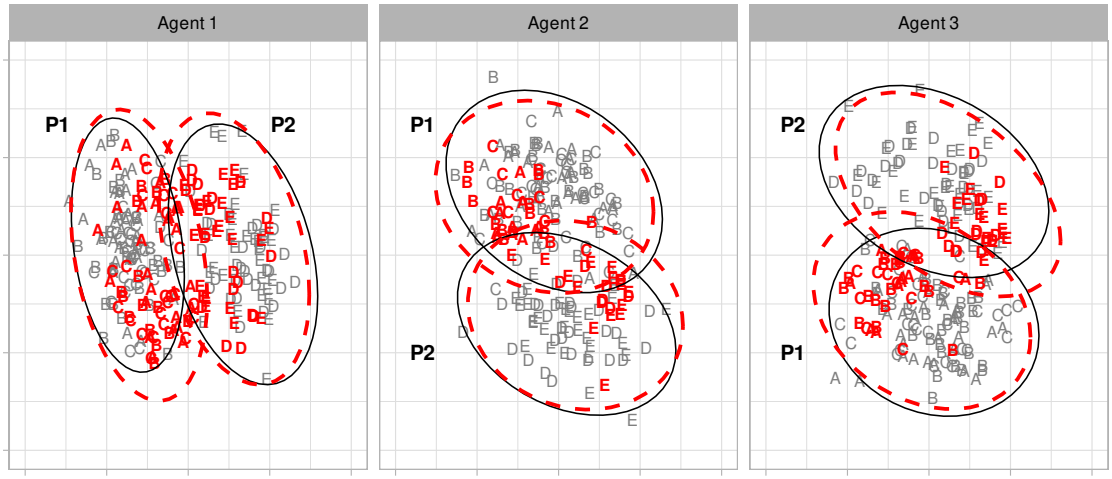
acoustically positioned.



(a)



(b)

Figure 6: A distribution of five words (A-E: 30 per word and per agent) in a two-dimensional space that were categorised as two phonological classes P1 and P2. Grey letters and black ellipses denote the initial conditions prior to the ABM simulation. Red letters and dashed ellipses are the conditions after interaction.

The evidence that such an inconsistency contributes to the merger is further illustrated in the simulations in Figure 6. In both 6a and 6b, the letters A-E represent different words. Figure 6a shows the outcome when the positioning of words in an acoustic space is consistent across agents. In this case, there are two phonological categories, P1 and P2, that map to words P1 $\in$ {A, B, C} and P2 $\in$ {D, E}. Moreover, P1 is positioned to the left of P2 on the x-axis for all three agents. The outcome of applying the ABM to these artificially created data in Figure 6a was the preservation of the phonological distinction between P1 and P2. The reason why the phonological separation was preserved was because the perceived, incoming acoustic exemplars that agents received from each other were usually positioned within the same phonological category: for example, agent 1 received exemplars of A, B, C from agents 2 and 3 that typically fell somewhere around the centre of agent 1's acoustic space in which A, B, C were located (i.e. near the centre of P1). Consequently, interaction typically preserved the separation between the two phonological categories, even though, as Figure 6a shows, there was a certain degree of overlap between them.

The simulation in Figure 6b shows the outcome when there is an inconsistent relationship across agents in how

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words are positioned in an acoustic space. In Figure 6b, there were the same two phonological categories consisting of the same sets of words and the degree of overlap between P1 and P2 was also much the same as in Figure 6a. But whereas in Figure 6b P1 and P2 are distinguished by the x-axis parameter for agent 1, for agents 2 and 3 it is the y-axis parameter that separates them; in addition, P1 is above P2 on the y-axis for agent 2, but below P2 for agent 3. Interaction between the agents in Figure 6b caused an increase in the overlap between the two phonological categories. This happened because perceived exemplars from other agents were typically positioned at the periphery of a phonological category. For example, agent 1 in Figure 6b received many incoming A, B, C words at the top of the P1 ellipse from agent 2 and at the bottom of the P1 ellipse from agent 3. Because these perceived tokens fell within agent 1's P1 ellipse, they were memorised. Because they were at the ellipse periphery, the effect of memorisation was to broaden the ellipse. A broadening of the ellipses, which happens for the same reason to both P1 and P2 and for all three agents, is the pre-condition for a merger to be possible. Thus, to summarise: an inconsistent positioning of the same words in an acoustic space across agents (Figure 6b) often leads to a merger because such inconsistency causes a broadening of phonological categories during interaction.

This type of inconsistency between speakers has been observed for some types of so-called marginal phonological relationships (Hall, 2013; Ladd, 2006; Nadeu & Renwick, 2016; Scobbie & Stuart-Smith, 2008), that have been characterised as intermediate between unambiguous phonological contrasts and neutralised forms. An example of an intermediate phonological relationship concerns the status of mid vowels in many Romance languages which have been neutralised within the front /e, ɛ/ and within the back /o, ɔ/ vowel sets (Recasens & Espinosa, 2009). In some varieties of Catalan by contrast, the distinction is maintained (Nadeu & Renwick, 2016). However, these vowel pairs differ from many other types of phonological contrast, not only because the functional load due to the distinction is small, but also because speakers even of the same dialect often disagree about how /e/ vs. /ɛ/ or /o/ vs. /ɔ/ divide up the vocabulary. A similar finding is reported in Renwick and Ladd (2016, p. 23) for Italian who show "significant lexical variability between speakers in the choice of mid vowel in stressed syllables" and that "speakers' productions do not always match their judgment about which vowel they use in which word". Just this type of inconsistency in how phonological categories map to lexical items across speakers is predicted by the simulation in Figure 6b to be a prior stage along the path towards complete neutralisation that is found in many other varieties of Romance languages.

Finally, the agent-based model developed in this study and the associated model of sound change are founded upon the idea that the phonology that links acoustics and the lexicon is stochastic and speaker-specific. Because of feedback connections between different levels of the model (Hay et al., 2015; Pierrehumbert, 2001, 2002, 2003; Wedel, 2006, 2012), phonological categories are incrementally updated when new exemplars have been memorised. A speaker-specific phonology means that speakers even of the same spoken accent do not necessarily carve up the lexicon into identical phonological categories. For the reasons outlined above, this speaker-specific phonology is one of the factors that can contribute to sound change. This view that phonology is stochastic and can vary between speakers is predicted by episodic models of speech (Beckner et al., 2009; Goldinger, 1996; Johnson, 2006; Pierrehumbert, 2003, 2006). This is because phonology emerges as a stochastic generalisation over lexical items which are in turn formed from memorised speech signals: these collections of exemplars are necessarily idiosyncratic because their accumulation is governed by the stochastic nature of a person's interlocutors and frequency of interaction. This is also consistent with many emergentist views of language (Bybee, 2002; Bybee & Good, 2008; Lindblom, 1998; Little et al., 2017; Studdert-Kennedy, 1998) in which the phonology of a population of speakers is shaped and updated as a consequence of spoken interactions between individuals that form part of the population (e.g. Beckner et al., 2009; Kirby & Sonderegger, 2013).

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Competing Interests

The authors have no competing interests to declare.

Appendices

A Gaussian Mixture Models-based clustering and classification

A Gaussian Mixture Model (GMM) is a parametric probability density function represented as a weighted sum of Gaussian component densities (Reynolds, 2009). Its general form is:

$$f(x|\lambda) = \sum_{j=1}^{G} w_j \cdot g(x|\mu_j, \Sigma_j), \tag{5}$$

where x is a continuous-valued d-dimensional vector, G is the number of components, $g(\cdot)$ is the multidimensional Gaussian density function and $\lambda = \{w_j, \mu_j, \Sigma_j\}$, $j = 1, \ldots, G$ indicates the set of weights w_j satisfying $\sum_{j=1}^{G} w_j = 1$, the d-dimensional mean vectors μ_j and the $d \times d$-dimensional covariance matrices Σ_j.

Given a set of N d-dimensional data vectors $\{x_i\}$, $i = 1, \ldots, N$, and fixing the number of components G, a model like Eq. (5) can be obtained by maximum likelihood (ML) estimation, usually applying the expectation-maximisation (EM) algorithm, as there is no closed-form solution for the maximisation of Eq. (5) with respect to λ. Often constraints are applied on the structure of Σ_j, e.g. diagonal, or parameters are tied across components, e.g. $\Sigma_j = \Sigma, \forall j$. The number of components can be estimated by first estimating several candidate models like Eq. (5), each one with a different number of components G, and then applying the Bayesian information criterion (BIC), or alternative IC methods, to select the best model.

Eq. (5) can be used as a clustering model for the data vectors $\{x_i\}$ that were used to estimate its parameters. By interpreting the mixture components as clusters, any vector x, from the training dataset or otherwise, can be assigned to a cluster j by applying the maximum a posteriori (MAP) criterion:

$$\mathit{Cluster}(x) = \arg\max_j \frac{w_j \cdot g(x|\mu_j, \Sigma_j)}{\sum_{k=1}^{G} w_k \cdot g(x|\mu_k, \Sigma_k)} = \arg\max_j w_j \cdot g(x|\mu_j, \Sigma_j). \tag{6}$$

When information on class membership of the data is available, i.e. the vectors x_i are paired with corresponding class labels $y_i \in \{1, \ldots, K\}$, where K is the number of classes, it is possible to use the form Eq. (5) as building block for a hierarchical classification model (Mixture Discriminant Analysis, MDA, Fraley & Raftery, 2002). MDA is a generalisation of Linear and Quadratic Discriminant Analysis (LDA, QDA) that allows the density of each class to be of the form of Eq. (5). Class membership y for a new vector x is obtained by applying Bayes's rule:

$$\mathit{Class}(x) = \arg\max_y Pr(x \in y) = \arg\max_y \frac{\tau_y \cdot f_y(x|\lambda_y, G_y)}{\sum_{k=1}^{K} \tau_k \cdot f_k(x|\lambda_k, G_k)} = \arg\max_y \tau_y \cdot f_y(x|\lambda_y, G_y), \tag{7}$$

where $\{\tau_y\}, y \in \{1, \ldots, K\}$ are the proportions of members of class y in the training set, $f_y(\cdot)$ are functions of the form of Eq. (5), each having in general a different parameter set λ_y and number of components G_y.

A.1 Implementation in the ABM

In the ABM code base, both GMM clustering and MDA classification are implemented using the R package *mclust* (Scrucca et al., 2016). The number of acoustic components, as well as a suitable set of constraints on the parameter set λ, are determined by the BIC criterion as implemented by the function `Mclust`. The MDA model is estimated using function `MclustDA`, where the number of components of each phonemic class is fixed according to the mapping obtained from the NMF procedure (see appendix C). For example, if clusters a_1 and a_2 map to sub-phoneme p_1 and a_3 maps to p_2, then the number of components for p_1 and p_2 are fixed to 2 and 1, respectively. This helps to preserve the original purely acoustic-based structure expressed by the GMM clustering model.

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B Non-negative Matrix Factorisation

Non-negative Matrix Factorisation (NMF, Seung & Lee, 2001) is a family of algorithms that given a $m \times n$ input matrix C with non-negative entries determines an approximation of it in the form of a product of two matrices, W and H, also with non-negative entries. Formally:

$$C \approx W \cdot H, \tag{8}$$

where W has dimension $m \times r$, H has dimension $r \times n$, and usually $r < \min(m, n)$. Once r is fixed, Eq. (8) is solved by minimising the difference between C and its approximation $W \cdot H$, i.e. W and H are the solution to:

$$\min_{W,H} \| C - W \cdot H \|_F^2, \text{ such that } W, H \geq 0, \tag{9}$$

where $\| \cdot \|_F$ is the Frobenius norm, the analogous of Euclidean norm for matrices.

In applications, each of the n columns c_j of C represents an observation or feature vector, where the m elements of c_j represent counts, frequencies, energies or other non-negative quantities. For example, in text mining the n columns of C represent documents and the m rows word counts or frequencies (Shahnaz et al., 2006). As the number of words $\times$ documents can be in the thousands $\times$ millions, it is convenient to find a compact approximation of C in the form of Eq. (8) with r as small as possible, say in the tens. Similarly to PCA, the columns of W can be seen as bases or principal components capturing the fundamental traits of the columns of C (though the columns of W are not constrained to be orthogonal), while each column h_j of H determines the linear combination of columns of W that best reconstructs a corresponding input column c_j. Differently from PCA, the non-negativity of W and H in Eq. (8) allows to interpret the composition of columns of W as strictly additive contribution from each basis in the quantities determined by the positive coefficients in H, where no contribution can undo (i.e. subtract) parts of the others. For example, in text mining the r columns of W are interpreted as topics, and each document c_j, codified by the number of occurrences of m selected words (i.e. bag of words representation), is a weighted sum of topics, the weights being the elements of column h_j of H.

A further step in the interpretation of Eq. (8) is achieved whenever W and H are sufficiently sparse, in which case the columns of H tend to ideally have only one large entry, which identifies only one column of W. This means that each of the n input column vectors of C can be approximated by just a coefficient multiplying one of the $r < n$ columns w_k of W, i.e. $c_j \approx H(k, j) \cdot w_k$ when $H(k, j) \gg H(i, j)$, $\forall i \neq k$. The particular column w_k approximating a given input vector c_j can be interpreted as its centroid or prototype, analogously to other prototype-based clustering algorithms, e.g. k-means (see Ding et al., 2005, for a formal statement). This interpretation allows to utilise NMF as a clustering algorithm, with r being the number of clusters. In order to improve the performance of NMF as clustering method, several sparse solutions have been proposed (e.g. Kim & Park, 2008; Pascual-Montano et al., 2006). As for most clustering algorithms, the optimal number of clusters r has to be determined by applying a model selection criterion external to the algorithm itself.

B.1 Implementation in the ABM

In the ABM code base, NMF clustering is implemented using the the R package *NMF* (Gaujoux & Seoighe, 2010). In particular, the NMF algorithm version by Pascual-Montano et al. (2006) is selected, which produces sparser solutions (setting method option to 'nsNMF' in the `nmf` command). The cluster assignment is determined by taking the arg max of each column of matrix H.

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Symbols	Meaning
x	an acoustic vector, e.g. $[1.2, -0.4, 0.88]^T$, when the acoustic space is 3-dimensional
d	number of dimensions of the acoustic space
w	a word type and a specific position within the word, e.g. *food*
e	an exemplar, i.e. a tuple (x, w)
$x(e)$	acoustic vector of exemplar e
$w(e)$	word type of exemplar e
N_e	number of exemplars in an agent's memory
N_a	number of acoustic clusters
N_p	number of sub-phonemic classes
N_w	number of word types
a	an acoustic cluster, determined by probabilistic membership to a Gaussian mixture component
p	a sub-phonemic class, determined by applying NMF to acoustic clusters
C_a	$N_w \times N_a$ matrix of exemplar counts, $C_a(j,k)$ is the count of exemplars of word w_j that belong to acoustic cluster a_k
C_p	$N_w \times N_p$ matrix of exemplar counts, $C_p(j,k)$ is the count of exemplars of word w_j that belong to sub-phonemic class p_k; these counts allow impurities. The sub-phonemic class of word type w_j is determined by picking $\arg\max_k C_p(j,k)$
$a(x(e)) = a(e)$	acoustic cluster associated to an acoustic vector
$p(a(e)) = p(a)$	sub-phonemic class associated to an acoustic cluster
$p(w(e)) = p(w)$	sub-phonemic class associated to a (position within a) word
Relations	**Comment**
$1 \leq N_p \leq N_a$	highest number of sub-phonemic classes occurs when each acoustic cluster constitutes a phonological class
$p(a(e)) \neq p(w(e))$	if an exemplar is 'impure', the sub-phonemic class associated to its location in the acoustic space is different from that associated to the word it belongs to

Table 2: Symbols used in Appendix C.

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C Implementation of acoustic and sub-phonological mental representations

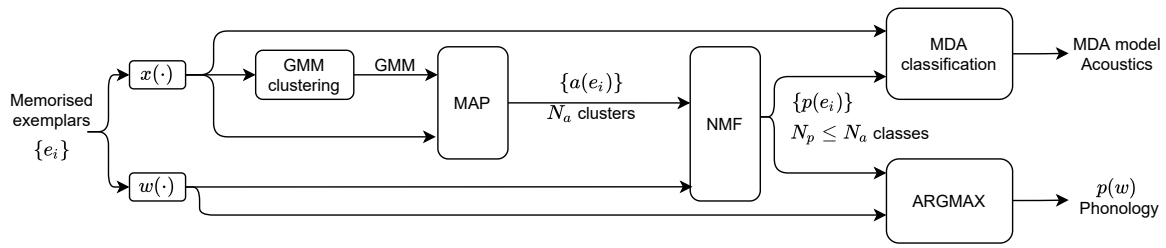

Figure 7: General scheme of the implementation of acoustic and sub-phonemic representations in the ABM.

C.1 Overview

The fundamental unit of an agent's memory is the exemplar, sometimes also called token in the main text. An exemplar e contains both the acoustic representation of an exemplar, i.e. an acoustic vector x, as well as the word type w and, implicitly, the position within the word associated to the exemplar, e.g. *food*. Each agent periodically re-estimates its acoustic and sub-phonemic models as follows. First, a GMM clustering model as in Eq. (5) is estimated from the acoustic representation of all the $N = N_e$ exemplars stored in memory, i.e.

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the set $\{x(e_i)\}$, $i \in \{1, \ldots, N_e\}$, disregarding any information on sub-phonemic classes established in previous estimations. The number of acoustic clusters $N_a = G$ is estimated by BIC. By applying Eq. (6) each exemplar is assigned to an acoustic cluster $a \in \{1, \ldots, N_a\}$ corresponding to a single Gaussian component. The acoustic cluster membership information associated to each exemplar $\{a(e_i)\}$, but *not* the corresponding acoustic representation $\{x(e_i)\}$, is fed to the NMF-based procedure (section C.2), which combines it with the word membership $\{w(e_i)\}$ to determine which acoustic clusters shall be joined together, i.e. a mapping from the acoustic labels $\{1, \ldots, N_a\}$ to the sub-phonemic labels $\{1, \ldots, N_p\}$, $N_p \leq N_a$. This mapping allows to indirectly associate each exemplar with a sub-phonemic class, i.e. $p(a(e_i))$. Using these class assignments, a MDA classification model is estimated, where the number of classes coincides with the number of sub-phonemic classes estimated by NMF, i.e. $K = N_p$ in Eq. (7). Such model is mainly responsible for the implementation of the relative criterion for exemplar memorisation (section 2.2), namely a new exemplar x, which according to phonology, i.e. via word membership $p(w(e_i))$, belongs to phoneme y, is memorised only if Eq. (7) also yields y as class membership. Note that the MDA model is based on class memberships obtained via the chain $p(a(e_i))$, which in general does not yield the same result as the one that uses word membership, i.e. $p(w(e_i))$. The reason is that acoustic clusters contain impurities, i.e. exemplars that do not belong to the sub-phonemic class that is present as majority in a given acoustic cluster. We prefer to use this 'impure' association to build the classification model because this represents the acoustic counterpart of the sub-phonemic organisation of words, hence should be based on acoustic similarity rather than on pure word membership.

C.2 How NMF identifies sub-phonemic classes

The input to NMF are (i) the set of acoustic cluster memberships $\{a(e_i)\}$, $i \in \{1, \ldots, N_e\}$, obtained by applying MAP as in Eq. (6) to the acoustic exemplars $\{x(e_i)\}$, and (ii) the set of word memberships of each exemplar, $\{w(e_i)\}$. These two pieces of information are combined into a $N_w \times N_a$ matrix of counts C_a, whose (j, k) element is the count of exemplars of word w_j that belong to acoustic cluster a_k, i.e. $C_a(j, k) = |\{i : w(e_i) = w_j, a(e_i) = a_k\}|$. This matrix is decomposed into the product of two matrices using NMF as in Eq. (8), i.e. $C_a \approx W \cdot H$, where the rank of the approximation corresponds to the number of sub-phonemic classes ($r = N_p$) and it is empirically determined as explained later on. Matrix H is further processed to obtain an indicator matrix $\widetilde{H}$, where each column h_k of H is substituted with a column of zeros and ones, with a single 1 at the position corresponding to the maximum element of h_k and zeros elsewhere. This is then used to recompute word counts according to the new N_p sub-phonemic clusters:

$$C_p = C_a \cdot \widetilde{H}^T, \tag{10}$$

where C_p is a $N_w \times N_p$ count matrix and ^{T} is matrix transposition. An example taken from the SSBE dataset (section 3) is illustrated as follows. One of the 22 agents at simulation start has the acoustic cluster counts C_a shown in Table 3a, where we see that this agent divides its acoustic space into $N_a = 5$ clusters, a_1 to a_5, corresponding to five Gaussian mixture components. Applying Eq. (8) to C_a, where we fix the rank to $r = N_p = 3$, we obtain:

$$\underbrace{\begin{bmatrix} 0 & 0 & 2 & 7 & 0 \\ 0 & 3 & 0 & 0 & 7 \\ 9 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 10 & 0 \\ 0 & 9 & 0 & 0 & 1 \\ 8 & 0 & 2 & 0 & 0 \\ 0 & 7 & 0 & 0 & 3 \\ 9 & 0 & 1 & 0 & 0 \\ 0 & 9 & 0 & 0 & 1 \\ 3 & 0 & 6 & 1 & 0 \\ 0 & 0 & 0 & 10 & 0 \end{bmatrix}}_{C_a} \approx \underbrace{\begin{bmatrix} 0 & 0 & 0.31 \\ 0 & 0.25 & 0 \\ 0.25 & 0 & 0 \\ 0 & 0 & 0.34 \\ 0 & 0.25 & 0 \\ 0.25 & 0 & 0 \\ 0 & 0.25 & 0 \\ 0.25 & 0 & 0 \\ 0 & 0.25 & 0 \\ 0.25 & 0 & 0 \\ 0 & 0 & 0.34 \end{bmatrix}}_{W} \times \underbrace{\begin{bmatrix} 29 & 0 & 12 & 0 & 0 \\ 0 & 28 & 0 & 0 & 12 \\ 0 & 0 & 0 & 28 & 0 \end{bmatrix}}_{H}, \tag{11}$$

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where we note that the NMF solution is indeed sparse. For example, the first column of C_a is approximated by the first column of W multiplied by 29, the second column of C_a by the second column of W multiplied by 28, etc., each column of C_a is approximated by just one column of W[13]. This provides the foundation for using the NMF expansion as a clustering criterion, namely columns of C_a approximated by the same column of W, such as columns 1 and 3, belong together because they are approximately proportional, which in our application means that they roughly contain the same word types. By observing C_a in Table 3a we can see that this is the case for acoustic clusters a_1 and a_3, which makes them good candidates for a merge. The merge of acoustic categories is obtained by constructing the indicator matrix $\widetilde{H}$ from H:

$$\widetilde{H} = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix},$$

the rows of which indicate with 1s the columns of C_a which should be merged (summed) into columns of C_p, e.g. the first row indicates that the first and third column of C_a will be summed together to become the first column of C_p. This operation is formalised by Eq. (10), the result of which is Table 3b.

	Acoustic clusters				
word	a_1	a_2	a_3	a_4	a_5
cooed	0	0	2	7	0
feed	0	3	0	0	7
feud	9	0	1	0	0
food	0	0	0	10	0
heed	0	9	0	0	1
hewed	8	0	2	0	0
keyed	0	7	0	0	3
queued	9	0	1	0	0
seep	0	9	0	0	1
soup	3	0	6	1	0
who'd	0	0	0	10	0

(a) C_a

	Sub-phonemic clusters		
word	p_1	p_2	p_3
cooed	2	0	**7**
feed	0	**10**	0
feud	**10**	0	0
food	0	0	**10**
heed	0	**10**	0
hewed	**10**	0	0
keyed	0	**10**	0
queued	**10**	0	0
seep	0	**10**	0
soup	**9**	0	1
who'd	0	0	**10**

(b) C_p

	Sub-phonemic classes
word	
cooed	p_3
feed	p_2
feud	p_1
food	p_3
heed	p_2
hewed	p_1
keyed	p_2
queued	p_1
seep	p_2
soup	p_1
who'd	p_3

(c) $p(w)$

Table 3: exemplar counts by word type (rows) and by acoustic cluster (columns, 3a) or by phonemic cluster (columns, 3b). This is a snapshot of the memory of one of the agents from the /u/-fronting case reported in section 3 taken at simulation start. In 3b boldface figures indicate which sub-phonemic cluster contains the majority of exemplars for each word type, which determines the assignment of words to sub-phonemic classes $p(w)$ in 3c.

The result in Eq. (10) defines the mapping from acoustic clusters to sub-phonemic clusters $p(a(e_i))$, which is not yet a definition based on word types, rather on word exemplars. For example, according to Table 3b two

[13]The reader unfamiliar with matrix multiplication may consult any text on linear algebra.

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exemplars of *cooed* belong to sub-phonemic class p_1 and seven to p_3. To obtain a many-to-one mapping from word types (and position within word) to sub-phonemic classes we assign each word w to the sub-phonemic cluster with highest exemplar count, i.e.

$$p(w) = \arg\max_p C_p(w, p). \tag{12}$$

For example, $p(\text{cooed}) = p_3$, $p(\text{feed}) = p_2$, etc., as indicated by boldface figures in Table 3b and correspondingly by labels in Table 3c. We call the entities defined by the mapping in Eq. (12) sub-phonemic *categories* or *classes*, to distinguish them from the sub-phonemic *clusters* identified by exemplar counts in C_p. This distinction is crucial as it brings along the concept of cluster *purity*, which plays a central role in determining the optimal number of sub-phonemic clusters, as explained below.

Purity is defined for each cluster as the fraction of exemplars that belong to that cluster, according to the majority rule defined in Eq. (12), divided by its exemplar count. Purity of the whole clustering solution is defined as the minimum purity value among the clusters (other definitions are possible, e.g. taking the average instead of the minimum, cf. Manning et al. (2008)). Formally:

$$\mathit{Purity} = \min_k \left(\frac{|\{i : p(a(e_i)) = p(w(e_i)) = p_k\}|}{|\{i : p(a(e_i)) = p_k\}|} \right) \tag{13}$$

To illustrate, the purity of each cluster in Table 3b is the sum of boldface counts divided by all counts in each column, e.g. $\mathit{Purity}(p_1) = \frac{39}{41} = 0.951$, which is the smallest value among the three ($\mathit{Purity}(p_2) = 1.0$ and $\mathit{Purity}(p_3) = \frac{27}{28} = 0.964$), so also the purity of the whole clustering solution according to Eq. (13). The optimal number of sub-phonemic clusters N_p is determined empirically by running several instances of NMF for each value of N_p from 2 to N_a, and choose the result that on average reaches the best trade-off between a high number of clusters and a high purity. This criterion in general disfavours the two extreme solutions, i.e. $N_p = 1$ and $N_p = N_a$. On the one hand, the highest number of clusters $N_p = N_a$ is probably not the solution with the highest purity; on the other hand, the trivial solution $N_p = 1$, which always yields a purity of 1, has also the lowest number of clusters. The trade-off is regulated by a purity threshold θ_{purity}, which has to be set by the user. The optimal N_p is chosen to be the largest among those corresponding to an average purity above θ_{purity}. Note that the solution $N_p = 1$ can be chosen, provided that purity at $N_p = 2$ is below θ_{purity}. To illustrate, Table 4 shows how purity was computed for the example in Table 3 with Eq. (13) five times for each candidate number of sub-phonemic clusters N_p, from 2 to $N_a = 5$, and then the optimal value $N_p = 3$ was chosen as the highest among the solutions above $\theta_{\text{purity}} = 0.75$.

N_p	Purity estimates					Average
2	0.95	0.95	1.0	1.0	0.9	0.96
3	0.95	0.95	0.95	0.95	0.95	**0.95**
4	0.58	0.50	0.50	0.50	0.50	0.52
5	0.50	-	0.50	0.50	0.50	-

Table 4: Five computations of purity and their average value (columns) for different number of sub-phonemic clusters (rows). The optimal solution is $N_p = 3$, as its average is the one with the highest N_p among the solutions with average purity above $\theta_{\text{purity}} = 0.75$. The stochasticity of NMF allows for different solutions starting from the same input (rows). Missing values can occur whenever one or more clusters are empty, and such solutions are never chosen, as we consider them unstable.

D SMOTE

D.1 Algorithm

Synthetic Minority Over-sampling TEchnique (SMOTE, Chawla et al., 2002) is a non-parametric resampling algorithm used to mitigate the negative impact of imbalanced datasets on supervised classification. SMOTE has been applied to several classification problems (see Fernandez et al., 2018, for a survey), its performance and theoretical properties have been evaluated (Blagus & Lusa, 2013), and a number of modifications to the original

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algorithm have been proposed (Fernandez et al., 2018; Leevy et al., 2018). In its basic form, SMOTE generates extra artificial data for the under-represented class(es) in such a way as to preserve their original statistical properties. For example, if class A contains only 10 data points, while class B has 100, SMOTE can be applied to class A to top it up to the level of class B by generating 90 artificial data points. The generative process only makes use of the existing data points from the under-represented class. In other words, SMOTE does not require the existence of a numerous class to generate extra data for a less numerous class. As a consequence, SMOTE can be applied more generally to increase the number of data points, e.g. when data are insufficient (though possibly balanced) for a given task.

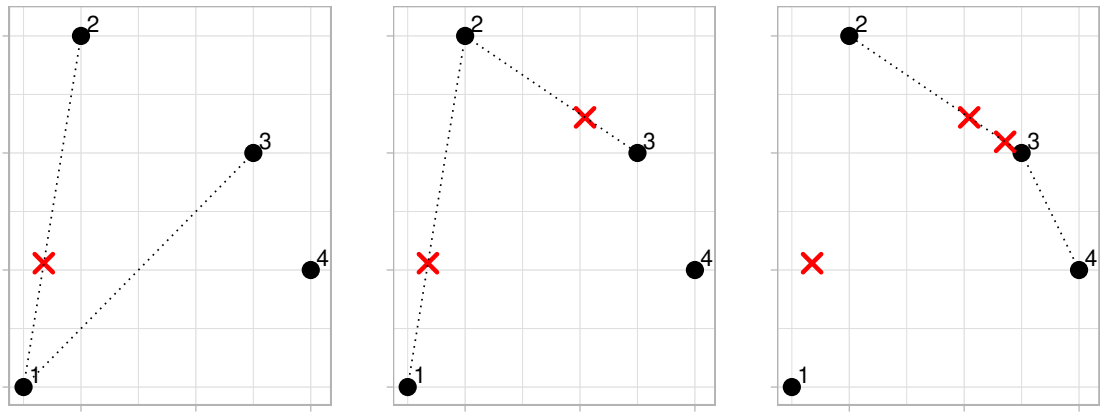

Figure 8: Example of application of SMOTE to a set of four points (numbered). Each panel shows the generation of one new point (red ×'s), in succession from left to right. In each panel, a pivot point (point 1, 2, and 3 from left to right) is connected by segments to its $k = 2$ nearest neighbours. A new point is generated at a random position on one of those segments. The new points are not used in the generation process, i.e. they do not become pivot points nor are they considered as neighbours.

The data generation mechanism of SMOTE is illustrated in Figure 8. Suppose there are n data points living in a d-dimensional space ($n = 4$ numbered points in a $d = 2$-dimensional space in Figure 8) and n_S new points need to be created. Each new data point (red ×'s in Figure 8) is generated by selecting a pivot point from the original n data points, connecting it to its k nearest neighbours ($k = 2$ in Figure 8), selecting one of such connecting segments at random and generating a new data point along that segment at a random location. The process is repeated n_S times, each time changing the pivot point. Figure 8 shows the generation of the first three points. Both k and n_S are parameters to be set by the user.

D.2 Implementation in the ABM

SMOTE has been employed in the ABM as a robustness measure for production. As recalled in section 2.1, the generation of a new acoustic token for speech production is implemented by first selecting a word class, then using all the exemplars of that word class available in the agent's memory to estimate a Gaussian distribution. The new token is produced by extracting one sample from that distribution. When the number of exemplars (data points) $N_e(w)$ of the target word class w is insufficient for a reliable estimation of a (multi-dimensional) Gaussian distribution, i.e. $N_e(w) < N_{\text{prod}}$, where N_{prod} is a user-defined threshold, $N_{\text{prod}} - N_e(w)$ extra exemplars for the target class are generated on the fly via SMOTE. After estimation, the extra exemplars are discarded. In case $N_e(w) < k + 1$, i.e. if the number of available exemplars $N_e(w)$ is not even sufficient to identify k nearest neighbours, $k - N_e(w) + 1$ exemplars in the acoustic proximity of the $N_e(w)$ target exemplars from other word classes, but belonging to the same sub-phonemic class $p(w)$ as the target word, are added to the set of original data points used by SMOTE to generate new points.

The pure application of the above strategy was found to be insufficient as a protection against instability, especially at the beginning of ABM simulations. This is the case when very few tokens from real speakers are available for the initialisation of agents, so much so that not only production but also perception becomes unreliable,

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as the estimation of Gaussian mixtures necessary for the identification of acoustic clusters (section 2.2.1) is also negatively affected by data scarcity. To mitigate the problem, a strategy to populate agent memories based on production, thus ultimately based on SMOTE, was introduced. A threshold $N_{\text{ex/word}}$ defining the minimum number of exemplars per word per agent available at all times throughout a simulation was introduced. For all cases not meeting such requirement at the beginning of a simulation, i.e. if $N_e(w) < N_{\text{ex/word}}$ for some word w, $N_{\text{ex/word}} - N_e(w)$ extra exemplars are generated by the same mechanism used for production and stored in the agent's memory. To exemplify, suppose $N_e(w) = 3$ and $N_{\text{ex/word}} = 10$ for a given agent and word w at simulation start. Then 7 extra exemplars are produced, each one applying the ordinary production algorithm based on the 3 original exemplars. If $N_e(w) < N_{\text{prod}}$, then SMOTE is applied each time, as explained above. Note that once a new exemplar is generated, it is not immediately made available as basis for the generation of the next ones. This is to avoid propagation of estimation errors. Only after all $N_{\text{ex/word}} - N_e(w)$ extra exemplars are generated, these are stored in memory and treated as original exemplars once the actual simulation (i.e. agent interactions) begins. To keep $N_e(w) \geq N_{\text{ex/word}}$ throughout the simulation, a constraint on the forgetting process was imposed such that no exemplar from a word class w can be forgotten (removed) if as a result the number of exemplars for w would go below the threshold $N_{\text{ex/word}}$ (cf. section 2.2.3). Note that $N_{\text{ex/word}}$ and N_{prod} are thresholds controlling different aspects, as the former is there to make sure that there is a minimum number of actual exemplars per word class in memory, which influences all aspects of the simulation, while the latter controls for the estimation basis for production only. In general $N_{\text{prod}} \geq N_{\text{ex/word}}$, for example if $N_{\text{ex/word}} = 10$ and $N_{\text{prod}} = 20$ it means that (i) $N_e(w) \geq 10$ for all w in all agents and (ii) SMOTE is applied in production whenever $10 \leq N_e(w) < 20$.

The SMOTE-related settings in the simulations presented in this work were fixed empirically to $N_{\text{prod}} = k + 1 = N_{\text{ex/word}} = 10$. This means in practice that no extra exemplars were generated at the beginning of the SSBE simulations (section 3), as they already met the minimum exemplar requirement, while for the NZE simulations (section 4) an aggressive initial re-population of agent memories was carried out, because the number of exemplars from real speakers per word per agent was usually only 1 or 2. In both cases, SMOTE was never applied during the simulations since $N_{\text{prod}} = N_{\text{ex/word}}$.

E GAMM-based analysis of results

Generalised Additive Mixed Models (GAMMs, Wood, 2017) were employed to analyse the acoustic aspect of ABM simulation results. The choice of GAMMs, as opposed to more conventional linear (mixed effects) models, is dictated by the need of testing a number of hypotheses on the evolution of formant track shape along simulation time. While agents do not exchange formant tracks directly, but rather acoustic vectors derived from them (cf. section 2), the acoustic realisations that the agents produce during a simulation need to be analysed according to what they actually correspond to in terms of speech signal. This is why the acoustic results from the ABM were first converted from PC scores back to formant tracks by applying FPCA-based reconstruction formulae (1a) and (1b), and then systematic changes in formant shape were assessed using GAMMs.

In order to assess the significance of specific hypotheses (e.g. F1 rises more in group A than in group B), the procedure in Wieling (2018) was applied. In particular, different GAMMs were estimated on the same data where the composition of so-called binary difference smooth terms varied. These terms are difference curves that can be used to test specific hypotheses (e.g. difference between F1 tracks in group A and B). Apart from that, all models are equivalent in terms of their predictions, which allowed to use only one of them to produce the plots in Figure 2 and 4. Separate GAMMs were estimated on F1 and on F2 tracks.

E.1 SSBE simulations

In section 3.1 hypotheses on the outcome of the simulations of SSBE sound change were stated, two of them pertain to the acoustics, which we repeat below for convenience:

1. Older agents /ju, u/ should shift more in the direction of younger agents' /ju, u/ than the other way round.
2. There should be no change to /i/.

Three pairs of GAMMs will be introduced in order to test the above hypotheses as well as a number of other auxiliary hypotheses that are necessary sanity checks on the simulation outcomes. GAMMs within the same pair

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have the same predictor structure but either F1 or F2 as response variable. The first pair of models (section E.1.1) reveals changes from baseline to post-run separately for each phoneme and for each speaker group. This allows to asses whether changes occurred in the expected direction, e.g. F2 tracks for /u/ should move upwards for older speakers and downwards for younger speakers. The second pair of models (section E.1.2) determines whether formants from older and younger speakers kept their the relative position throughout the simulation. Even if not explicitly stated in the hypotheses above, we need to make sure that changes in the expected direction did not overshoot, e.g. F2 tracks for /u/ should move upwards for older speakers but should not end up being higher than those of younger speakers. Finally, the third pair of models (section E.1.3) determines if the difference between the change in older and younger speakers has the expected sign, as required by the first hypothesis above. That is, if older speakers changed more than younger speakers, the difference between the two is positive.

In all GAMMs the response variable is the trajectory of either F1 or F2 in normalised sampled time. Fixed smooth factors are age group (`gr`, levels: Y = younger, O = older), state (`st`, levels: BL = baseline, PR = post-run), and canonical phoneme (`ph`, levels: /i, u, ju/). The random smooth effects are agent (22 levels) interacting with state and phoneme, word (11 levels) interacting with group and state, and run (10 levels) interacting with group, state, and phoneme. For each pair of GAMMs, first the model equation is provided, with emphasis on the role of the binary difference smooths, then results are reported.

E.1.1 Testing changes separately

The models to assess the change from baseline to post-run separately for phoneme and group have the following equation:

$$\mathrm{F}(\mathtt{t}_i) = \sum_{\substack{\mathtt{ph}\in\{/\mathrm{i},\ \mathrm{u},\ \mathrm{ju}/\} \\ \mathtt{gr}\in\{\mathrm{Y},\ \mathrm{O}\}}} f_{\mathtt{ph},\mathtt{gr}}(\mathtt{t}_i) \cdot I(\mathtt{ph}_i = \mathtt{ph} \cap \mathtt{gr}_i = \mathtt{gr}) + b_{\mathtt{ph},\mathrm{PR},\mathtt{gr}}(\mathtt{t}_i) \cdot I(\mathtt{ph}_i = \mathtt{ph} \cap \mathtt{st}_i = \mathrm{PR} \cap \mathtt{gr}_i = \mathtt{gr}) + \mathrm{r.t.} + \epsilon_i, \quad (14)$$

where the response F is either F1 or F2, i is the running index over samples, $\mathtt{t}$ is sampled normalised time, f and b are ordinary and binary smooths, respectively, $I()$ is the indicator function taking value 1 when its argument is true and 0 otherwise, r.t. indicates random smooth terms (described above), and ϵ is a residual error term. The structure of model (14) allows to interpret b terms as difference curves between post-run and baseline for a specific phoneme × group combination. For example, if $\mathtt{ph}_i = /\mathrm{u}/$ and $\mathtt{gr}_i = \mathrm{Y}$, model (14) reads (omitting r.t. and ϵ): $\mathrm{F}(\mathtt{t}_i) = f_{/\mathrm{u}/,\mathrm{Y}}(\mathtt{t}_i)$ at baseline and: $\mathrm{F}(\mathtt{t}_i) = f_{/\mathrm{u}/,\mathrm{Y}}(\mathtt{t}_i) + b_{/\mathrm{u}/,\mathrm{PR},\mathrm{Y}}(\mathtt{t}_i)$ at post-run, thus $b_{/\mathrm{u}/,\mathrm{PR},\mathrm{Y}}(\mathtt{t}_i)$ is the formant track difference between post-run and baseline for younger speakers' /u/.

Figure 9 shows the estimated binary difference smooth terms b from model (14) predicting F1 (bottom row) and F2 (top row). The only significant difference term for F1 is (hereafter omitting time dependency) $b_{/\mathrm{u}/,\mathrm{PR},\mathrm{O}}$ (bottom left black curve, $edf = 2.0$, $rdf = 2.0$, $F = 5.4$, $p < 0.01$), which is a small change in tilt of F1 for old speakers' /u/. In the model predicting F2, the difference terms for /i/ were not significant, which supports hypothesis 2. The difference terms for /u/ are both significant (top mid panel, $b_{/\mathrm{u}/,\mathrm{PR},\mathrm{O}}$ in black, $edf = 8.9$, $rdf = 8.9$, $F = 41.7$, $p < 0.001$; $b_{/\mathrm{u}/,\mathrm{PR},\mathrm{Y}}$ in green, $edf = 5.2$, $rdf = 5.4$, $F = 2.2$, $p < 0.05$) and in the direction implied by hypothesis 1, as F2 for /u/ moves upwards in older speakers and downwards, but less so, in younger speakers, which according to Figure 2 means that older and younger speakers' /u/ move towards each other. Figure 9 shows a larger change for older speakers' than for younger speakers' /u/, but the models in section E.1.3 are needed to assess that this difference in change is significant. Finally, the difference term for /ju/ (top right panel) is not significant for older but it is for younger speakers ($edf = 5.1$, $rdf = 5.2$, $F = 2.5$, $p < 0.05$), which speaks against hypothesis 1 for /ju/, although also in this case the models in section E.1.3 will give a definitive answer.

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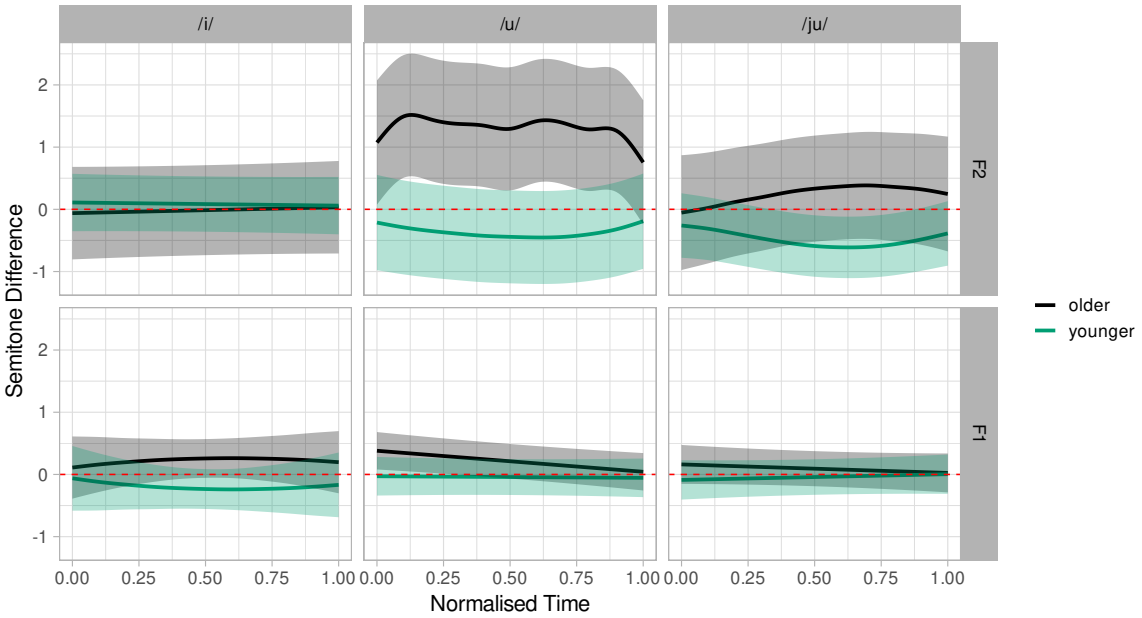

Figure 9: Binary difference smooth (b) estimates with 95% confidence bands from GAMM (14). Each panel shows $b_{\mathtt{ph},\mathrm{PR},\mathrm{O}}(\mathtt{t}_i)$ in black and $b_{\mathtt{ph},\mathrm{PR},\mathrm{Y}}(\mathtt{t}_i)$ in green, where ph is indicated by the column headings and the response variable by the rows. The horizontal axis at level zero is marked by red dashed lines.

E.1.2 Testing for no crossing

The models to assess the relative position of corresponding phonemes in younger and older speakers along the simulation have the following equation:

$$\mathrm{F}(\mathtt{t}_i) = \sum_{\substack{\mathtt{ph}\in\{/\mathrm{i},\ \mathrm{u},\ \mathrm{ju}/\} \\ \mathtt{st}\in\{\mathrm{BL},\ \mathrm{PR}\}}} f_{\mathtt{ph},\mathtt{st}}(\mathtt{t}_i) \cdot I(\mathtt{ph}_i = \mathtt{ph} \cap \mathtt{st}_i = \mathtt{st}) + b_{\mathtt{ph},\mathtt{st},\mathrm{Y}}(\mathtt{t}_i) \cdot I(\mathtt{ph}_i = \mathtt{ph} \cap \mathtt{st}_i = \mathtt{st} \cap \mathtt{gr}_i = \mathrm{Y}) + \mathrm{r.t.} + \epsilon_i, \quad (15)$$

with the same notation as in Eq. (14). Here b terms represent the difference between younger and older speakers' formants for a given phoneme × state combination.

Figure 10 shows the estimated binary difference smooth terms b from GAMM (15) predicting F1 (bottom row) and F2 (top row). All b terms are significant except for $b_{/\mathrm{i}/,\mathrm{PR},\mathrm{Y}}$ for F1. Details are reported in Table 5. The essential outcome from Figure 10 is that in all cases the $b_{\mathtt{ph},\mathrm{BL},\mathrm{Y}}$ and $b_{\mathtt{ph},\mathrm{PR},\mathrm{Y}}$ curves for the same phoneme

Response	Term	*edf*	*rdf*	*F*	*p*-value
F1	$b_{/\mathrm{i}/,\mathrm{BL},\mathrm{Y}}$	4.5	4.6	3.6	< 0.01
	$b_{/\mathrm{i}/,\mathrm{PR},\mathrm{Y}}$	2.5	2.7	1.4	n.s.
	$b_{/\mathrm{u}/,\mathrm{BL},\mathrm{Y}}$	2.0	2.0	25.9	< 0.001
	$b_{/\mathrm{u}/,\mathrm{PR},\mathrm{Y}}$	2.0	2.0	8.7	< 0.001
	$b_{/\mathrm{ju}/,\mathrm{BL},\mathrm{Y}}$	3.4	3.6	4.3	< 0.01
	$b_{/\mathrm{ju}/,\mathrm{PR},\mathrm{Y}}$	3.8	3.9	3.4	< 0.01
F2	$b_{/\mathrm{i}/,\mathrm{BL},\mathrm{Y}}$	5.4	5.5	2.9	< 0.01
	$b_{/\mathrm{i}/,\mathrm{PR},\mathrm{Y}}$	5.0	5.2	2.6	< 0.05
	$b_{/\mathrm{u}/,\mathrm{BL},\mathrm{Y}}$	9.3	9.3	85.4	< 0.001
	$b_{/\mathrm{u}/,\mathrm{PR},\mathrm{Y}}$	8.2	8.3	6.0	< 0.001
	$b_{/\mathrm{ju}/,\mathrm{BL},\mathrm{Y}}$	9.3	9.3	66.5	< 0.001
	$b_{/\mathrm{ju}/,\mathrm{PR},\mathrm{Y}}$	8.3	8.3	21.1	< 0.001

Table 5: Statistics for the binary difference smooth (b) estimates from GAMM (15).

and formant are on the same side of the horizontal axis at level zero. For example, F2 differences $b_{/\mathrm{u}/,\mathrm{BL},\mathrm{Y}}$ and $b_{/\mathrm{u}/,\mathrm{PR},\mathrm{Y}}$ (top mid panel) are both positive curves, which means that younger speakers' F2 tracks for /u/ are

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higher than those of older speakers both at baseline and at post-run, i.e. the simulation did not reverse their relative position.

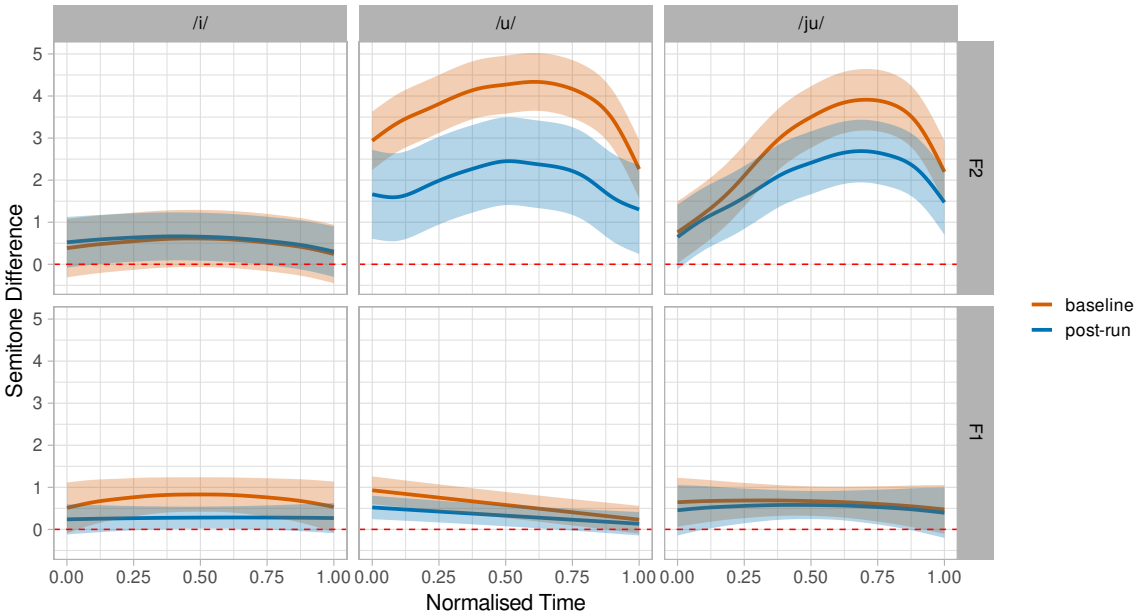

Figure 10: Binary difference smooth (b) estimates with 95% confidence bands from GAMM (15). Each panel shows $b_{\mathtt{ph},\mathrm{BL},\mathrm{Y}}(\mathtt{t}_i)$ in orange and $b_{\mathtt{ph},\mathrm{PR},\mathrm{Y}}(\mathtt{t}_i)$ in blue, where $\mathtt{ph}$ is indicated by the column headings and the response variable by the rows. The horizontal axis at level zero is marked by red dashed lines.

E.1.3 Testing differences of changes

A direct test of hypothesis 1 requires to take into account the relative directions of changes. For example, the change in F2 of /u/ is an upward shift for older and a downward shift for younger speakers, as shown in Figure 9 top mid panel. In order to test that the shift for older is larger in absolute value than the shift for younger speakers, a set of difference curves needs to be set up that allow to test for the following inequality:

$$\left(F2(\mathtt{t})\Big|_{/\mathrm{u}/,\mathrm{PR},\mathrm{O}} - F2(\mathtt{t})\Big|_{/\mathrm{u}/,\mathrm{BL},\mathrm{O}}\right) - \left(F2(\mathtt{t})\Big|_{/\mathrm{u}/,\mathrm{BL},\mathrm{Y}} - F2(\mathtt{t})\Big|_{/\mathrm{u}/,\mathrm{PR},\mathrm{Y}}\right) > 0, \tag{16}$$

where $x\Big|_y$ means x evaluated at y. In terms of GAMM (14), inequality (16) translates into $b_{/\mathrm{u}/,\mathrm{PR},\mathrm{O}} - (-b_{/\mathrm{u}/,\mathrm{PR},\mathrm{Y}}) > 0$, i.e. in Figure 9 top mid panel the difference between the black curve and the flipped image of the green curve with respect to the horizontal axis should be positive. The following GAMM serves the purpose, as it contains b terms that correspond to the expression on the left hand side of inequality (16) for each phoneme:

$$\begin{aligned}\mathrm{F}(\mathtt{t}_i) = \sum_{\substack{\mathtt{ph}\in\{/\mathrm{i},\ \mathrm{u},\ \mathrm{ju}/\}\\ \mathtt{gr}\in\{\mathrm{Y},\ \mathrm{O}\}}} f_{\mathtt{ph},\mathrm{PR}\oplus\mathtt{gr}}(\mathtt{t}_i)\cdot I(\mathtt{ph}_i = \mathtt{ph}\cap(\mathtt{st}_i = \mathrm{PR}\oplus\mathtt{gr}_i = \mathtt{gr}))+\\ \sum_{\mathtt{ph}\in\{/\mathrm{i},\ \mathrm{u},\ \mathrm{ju}/\}} b_{\mathtt{ph},\mathrm{O}}(\mathtt{t}_i)\cdot I(\mathtt{ph}_i = \mathtt{ph}\cap\mathtt{gr}_i = \mathrm{O})+\\ \sum_{\mathtt{ph}\in\{/\mathrm{i},\ \mathrm{u},\ \mathrm{ju}/\}} b_{\mathtt{ph},\mathrm{O},\mathrm{PR}\oplus\mathrm{Y}}(\mathtt{t}_i)\cdot I(\mathtt{ph}_i = \mathtt{ph}\cap\mathtt{st}_i = \mathrm{PR}\cap\mathtt{gr}_i = \mathrm{O})+\\ \text{r.t.} + \epsilon_i,\end{aligned} \tag{17}$$

where $\oplus$ indicates exclusive disjunction or XOR. To illustrate, Table 6 indicates for each term that appears in inequality (16) the corresponding sum of fixed smooths in GAMM (17). By substituting (17) into (16) most of the terms cancel each other out and we obtain: $b_{/\mathrm{u}/,\mathrm{O},\mathrm{PR}\oplus\mathrm{Y}} > 0$, i.e. a single smooth term that tests hypothesis 1 for F2 and /u/; a similar term is available for /ju/.

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Term in (16)	Terms in (17)
$F2(\mathtt{t})\big\vert_{/\mathrm{u}/,\mathrm{BL,Y}}$	$f_{/\mathrm{u}/,\mathrm{PR}\oplus\mathrm{Y}}$
$F2(\mathtt{t})\big\vert_{/\mathrm{u}/,\mathrm{PR,Y}}$	$f_{/\mathrm{u}/,\mathrm{PR}\oplus\mathrm{O}}$
$F2(\mathtt{t})\big\vert_{/\mathrm{u}/,\mathrm{BL,O}}$	$f_{/\mathrm{u}/,\mathrm{PR}\oplus\mathrm{O}} + b_{/\mathrm{u}/,\mathrm{O}}$
$F2(\mathtt{t})\big\vert_{/\mathrm{u}/,\mathrm{PR,O}}$	$f_{/\mathrm{u}/,\mathrm{PR}\oplus\mathrm{Y}} + b_{/\mathrm{u}/,\mathrm{O}} + b_{/\mathrm{u}/,\mathrm{O,PR}\oplus\mathrm{Y}}$

Table 6: Fixed smooths in GAMM (17) corresponding to the combinations of `ph`, `st` and `gr` occurring in inequality (16).

Figure 11 shows the estimated binary difference smooth terms $b_{\mathtt{ph},\mathrm{O},\mathrm{PR}\oplus\mathrm{O}}$ from GAMM (17) predicting F1 (bottom row) and F2 (top row). Only terms $b_{/\mathrm{u}/,\mathrm{O},\mathrm{PR}\oplus\mathrm{Y}}$ for both F1 and F2 are significant (F1, bottom mid panel, $edf = 2.0$, $rdf = 2.0$, $F = 3.1$, $p < 0.05$; F2, top mid panel, $edf = 8.5$, $rdf = 8.6$, $F = 10.7$, $p < 0.001$). The difference of changes in F1 of /u/ is quite small and does not lend itself to a clear interpretation. The main results concerning hypothesis 1 are that (i) the difference of changes in F2 of /u/ is significant and positive, as the curve in top mid panel of Figure 11 shows, which verifies inequality (16); (ii) the difference of changes in F2 of /ju/ is not significant, which is a null result for hypothesis 1 on /ju/ but not evidence against it, as the results from section E.1.1 seemed to suggest.

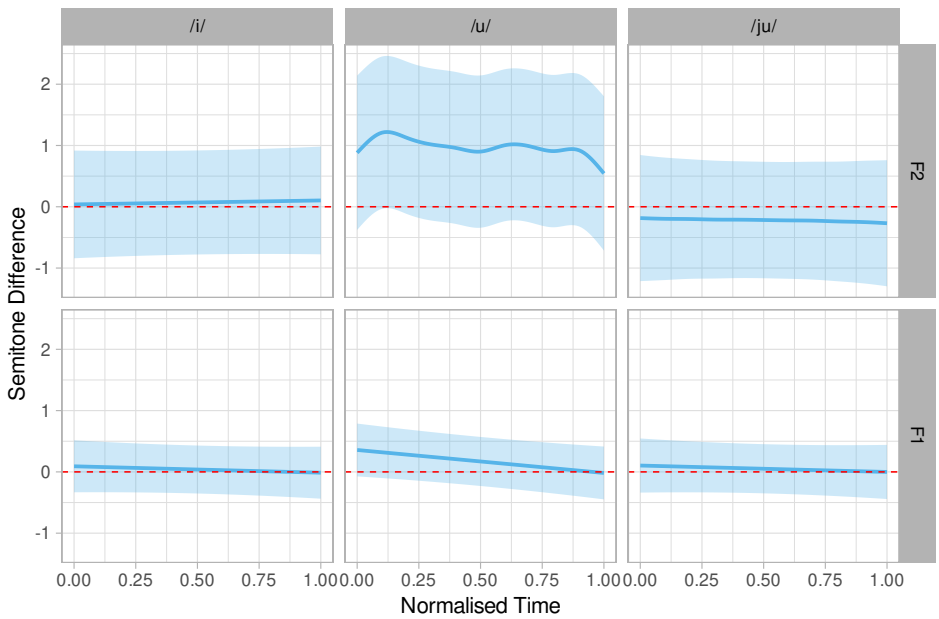

Figure 11: Binary difference smooth (b) estimates with 95% confidence bands from GAMM (17). Each panel shows $b_{\mathtt{ph},\mathrm{O},\mathrm{PR}\oplus\mathrm{Y}}$, where `ph` is indicated by the column headings and the response variable by the rows. The horizontal axis at level zero is marked by red dashed lines.

E.1.4 Summary

In summary, it was shown that older speakers shifted their /u/ towards that of younger speakers in the form of F2-raising and a slight change in tilt of F1. Younger speakers lowered the second formant in F2, but this change was smaller than the F2 change of older speakers' /u/ as shown in this section, which lends support to hypothesis 1. With regard to /ju/, the only significant change was that younger speakers lowered their F2; however, the GAMM in this section showed that this change was not significantly different from the no-change of older speakers' F2 for /ju/. Lastly, there were no changes to /i/ which supports hypothesis 2.

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E.2 NZE simulations

In section 4.1 hypotheses on the outcome of the simulations of NZE sound change were stated, two of them pertain to the acoustics, which we repeat below for convenience:

1. Older agents' /eə/ should shift more in the direction of younger agents' /eə/ than the other way round.
2. There should be no change to /ɪə/.

As these hypotheses clearly mirror those on SSBE simulations, the analysis will follow the same structure of section E.1. In particular, the same three pairs of GAMMs will be employed, with minor adjustments detailed as follows. In all GAMMs the response variable is the trajectory of either F1 or F2 in normalised sampled time. Fixed smooth factors are age group (`gr`, levels: Y = younger, O = older), state (`st`, levels: BL = baseline, PR = post-run), and canonical phoneme (`ph`, levels: /ɪə, eə/). The random smooth effects are agent (48 levels) interacting with state and phoneme, word (53 levels) interacting with group and state, and run (10 levels) interacting with group, state, and phoneme.

E.2.1 Testing changes separately

The models to assess the change from baseline to post-run separately for phoneme and group are the same as GAMM (14), with the difference that `ph` takes the two values /ɪə, eə/ and that r.t. are adjusted on the levels detailed above. Figure 12 shows the estimated binary difference smooth terms b predicting F1 (bottom row) and F2 (top row) using the same conventions as in Figure 9. Details are reported in Table 7. In the model predicting

Response	Term	*edf*	*rdf*	*F*	*p*-value
F1	$b_{/\text{ɪə}/,\text{PR,O}}$	2.5	2.6	5.1	< 0.01
	$b_{/\text{ɪə}/,\text{PR,Y}}$	2.0	2.0	1.6	n.s.
	$b_{/\text{eə}/,\text{PR,O}}$	5.0	5.2	4.0	< 0.01
	$b_{/\text{eə}/,\text{PR,Y}}$	2.0	2.0	2.8	$= 0.06$
F2	$b_{/\text{ɪə}/,\text{PR,O}}$	2.0	2.0	5.9	< 0.01
	$b_{/\text{ɪə}/,\text{PR,Y}}$	2.1	2.1	4.6	$= 0.01$
	$b_{/\text{eə}/,\text{PR,O}}$	2.0	2.0	2.4	n.s.
	$b_{/\text{eə}/,\text{PR,Y}}$	2.0	2.0	3.9	< 0.05

Table 7: Statistics for the binary difference smooth (b) estimates from GAMM (14) adapted to the NZE case.

F1 the change in /eə/ is significant for older ($b_{/\text{eə}/,\text{PR,O}}$) but not for younger speakers ($b_{/\text{eə}/,\text{PR,Y}}$), and the change is in the expected direction since $b_{/\text{eə}/,\text{PR,O}} < 0$, i.e. F1 in /eə/ tends to shift downwards along the simulation, especially in the first part of the diphthong (black curve, bottom right panel in Figure 12). These results support hypothesis 1. There is also a significant yet small change in tilt of /ɪə/ in older speakers, which is contrary to hypothesis 2. In the model predicting F2 all the changes are in terms of slight tilts (top panels in Figure 12), in particular younger speakers tend to tilt F2 of about the same amount both for /eə/ and /ɪə/.

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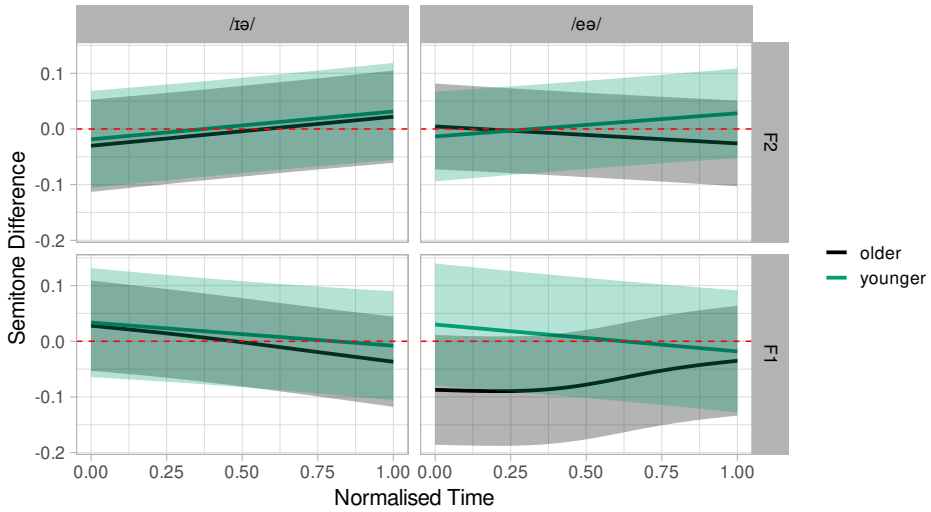

Figure 12: Binary difference smooth (b) estimates with 95% confidence bands from GAMM (14) adapted for the NZE case. Each panel shows $b_{\texttt{ph},\text{PR},\text{O}}(\texttt{t}_i)$ in black and $b_{\texttt{ph},\text{PR},\text{Y}}(\texttt{t}_i)$ in green, where `ph` is indicated by the column headings and the response variable by the rows. The horizontal axis at level zero is marked by red dashed lines.

E.2.2 Testing for no crossing

The models to assess the relative position of corresponding phonemes in younger and older speakers along the simulation are the same as GAMM (15), where the same adjustments from section E.2.1 apply. Figure 13 shows the estimated binary difference smooth terms b predicting F1 (bottom row) and F2 (top row) using the same conventions as in Figure 10. Only the terms $b_{/\text{eə}/,\text{BL},\text{O}}$ for both F1 and F2 are significant (F1, orange curve, bottom right panel, $edf = 5.8$, $rdf = 6.0$, $F = 6.4$, $p < 0.001$; F2, orange curve, top right panel, $edf = 4.8$, $rdf = 4.8$, $F = 5.7$, $p < 0.001$). First, these results do not support the hypothesis that the relative position of formants in younger and older speakers got reversed along the simulation, as no post-run difference is significant. Moreover, combining these results with those from section E.2.1 we can say that the absence of significant difference in /ɪə/ between younger and older speakers both at baseline and post-run mitigates the finding from section E.2.1 against hypothesis 2, i.e. there was a small change in /ɪə/ but this took place in all groups in the same way, which may not be related to the /ɪə, eə/ contrast but to the presence of the phonetic context pattern described in section 4.3.

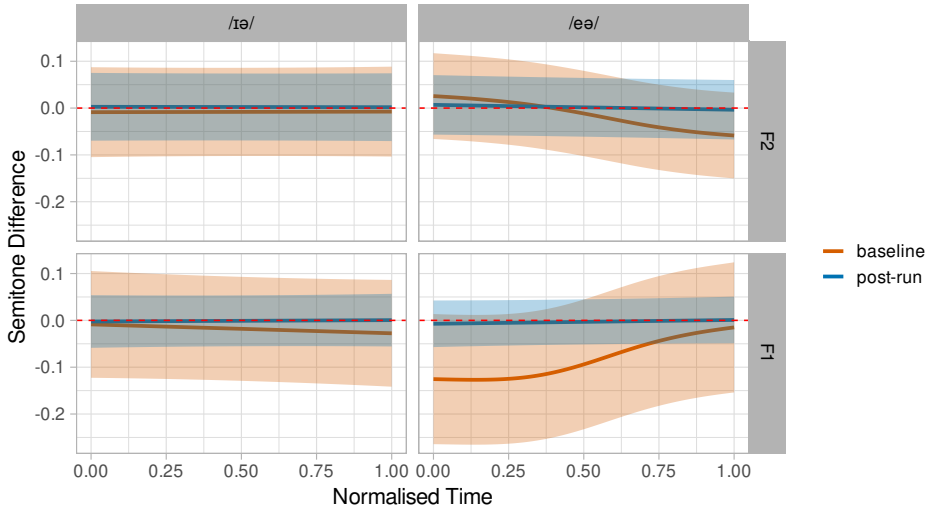

Figure 13: Binary difference smooth (b) estimates with 95% confidence bands from GAMM (15) adapted for the NZE case. Each panel shows $b_{\texttt{ph},\text{BL},\text{Y}}(\texttt{t}_i)$ in orange and $b_{\texttt{ph},\text{PR},\text{Y}}(\texttt{t}_i)$ in blue, where `ph` is indicated by the column headings and the response variable by the rows. The horizontal axis at level zero is marked by red dashed lines.

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E.2.3 Testing differences of changes

A direct test of hypothesis 1 requires to assess that the algebraic sum of the changes for older and for younger speakers' /eə/ (right panels in Figure 12) has the same sign of the older speakers' change. A pair of GAMMs with the same form as model (17) were estimated, applying the necessary adjustments as in sections E.2.1 and E.2.2. Figure 14 shows the estimated binary difference smooth terms $b_{\texttt{ph},\mathrm{O},\mathrm{PR}\oplus\mathrm{O}}$ predicting F1 (bottom row) and F2 (top row). The terms of interest for hypothesis 1 $b_{/\mathrm{eə}/,\mathrm{O},\mathrm{PR}\oplus\mathrm{Y}}$ are not significant (F1, bottom right panel, $edf = 2.0$, $rdf = 2.0$, $F = 0.3$, n.s.; F2, top right panel, $edf = 2.0$, $rdf = 2.0$, $F = 0.1$, n.s.).

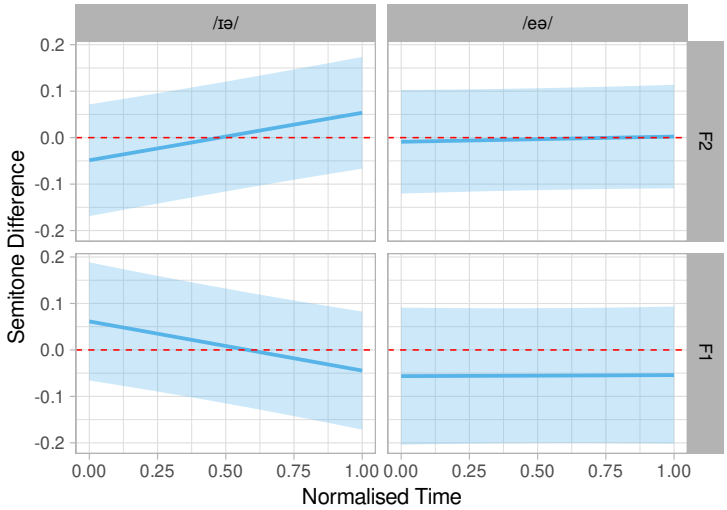

Figure 14: Binary difference smooth (b) estimates with 95% confidence bands from GAMM (17) adapted to the NZE case. Each panel shows $b_{\texttt{ph},\mathrm{O},\mathrm{PR}\oplus\mathrm{Y}}$, where `ph` is indicated by the column headings and the response variable by the rows. The horizontal axis at level zero is marked by red dashed lines.

E.2.4 Summary

In summary, it was shown that hypothesis 1 is partially confirmed, as the changes in F1 for older speakers' /eə/ are significant and in the expected direction, while no corresponding significant change occurred for younger speakers. However, the difference between these two changes was not significant. Hypothesis 2 is partially confirmed, as there were small changes in tilt for F1 and F2 in /ɪə/, but these occurred in younger and in older speakers equally, which suggests that these do not originate from the differences between the two groups canonical diphthongs at baseline.

F Further simulations on NZE

The results of the simulations presented in section 4.3 are characterised by the emergence of two patterns at the sub-phonological level: one is the /ɪə, eə/ merger which is in line with the direction of sound change, the other is the consolidation of an initially latent context-dependent binary subdivision. In order to assess the validity of the predictions from this particular simulation, a number of trivial explanations for its outcome need to be ruled out. Firstly, as there are more innovative than conservative agents in this simulation and most agents start with neutralised classes, we need to exclude that the merger occurred as a mere consequence of imbalance in the data. Secondly, we need to exclude that the stronger context-dependent pattern played a major role in neutralising the weaker /ɪə, eə/ subdivision. In this section we present results from further simulations aimed at ruling out these two explanations.

The merger of the centring diphthongs had taken hold in nearly all NZE speakers, although to different degrees. Most young speakers had fully neutralised the contrast between the two diphthongs; older speakers on the other hand were shown to differentiate between /ɪə/ and /eə/ at least some of the time. However, across the population the merger was relatively advanced so that there was arguably an imbalance in the input data in favour of the

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merger. Here we show that it is not a characteristic of the IP model that the majority always wins. We conducted a set of simulations based on agent populations where the proportion of conservative vs. innovative NZE speakers was manipulated. Each population counted 24 agents in total, the conservative and the innovative subsets were formed by (re-)sampling from the 12 NZE speakers whose sub-phonemes at simulation start most, resp. least agreed with the canonical phonemes /ɪə/ and /eə/. Seven proportions were considered, from all conservatives to all innovatives, and for each of them a set of five independent ABM runs was conducted using the same settings as in the original NZE simulations (section 4.2.2). As can be seen in Figure 15, most initial conditions of the NZE data tend towards a split of the diphthongs into canonical classes. For example, in the simulations based on a population composed of 12 conservative and 12 innovative agents the proportion of agents whose canonical agreement is at chance level decreases from around 35% at start to 10% at the end of the simulation, i.e. the majority of agents who started out with a merged variant of the diphthongs start to develop this contrast. The empirical, full dataset (in red) is one of the few conditions that produces a merger, as demonstrated by the fact that the proportion of agents whose canonical agreement is at chance level increases from around 30% to 65%.

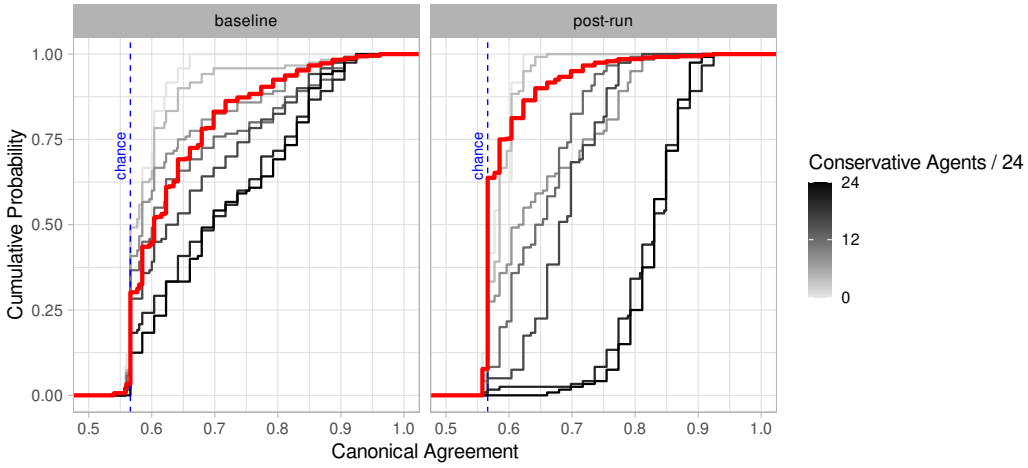

Figure 15: Empirical cumulative distributions of canonical agreement as computed by Eq. (2) at baseline and post-run for 7 batches of ABM simulations. Each batch is based on a population of 24 NZE agents with a different proportion of conservative vs. innovative speakers, indicated by the gray scale. For comparison, the original NZE simulation is indicated by a thicker red curve.

In the original NZE dataset, there was a coarticulatory influence of the phonetic context on the realisation of both diphthongs whereby those in pre-pausal position were more centralised than those followed by alveolar consonants. In the simulation then, this context-dependent variation was phonologised after the separation into canonical phonemes had ceased to exist (cf. Figure 5 and Table 1). However, the context-dependent pattern is not in any way causal to the merger. To verify that, simulations were run on manipulated versions of the original NZE agent population where random word label swaps were carried out on the exemplars stored in the agents' memory at the initialisation phase. The swaps were implemented always across the two context-dependent word groups described in Table 1 but within canonical /ɪə/ and /eə/. For example, in one agent all stored exemplars of *beard* (/ɪə/ and Group 1 in Table 1) were re-labelled as *sphere* (/ɪə/ and Group 2) and vice versa, in another agent *beard* was swapped with *fear* (/ɪə/ and Group 2), in another agent *beard* kept its original labelling, and so on. Three batches of simulations were set up, each consisting of five parallel runs, where 4, 8, or 12 word type swaps were applied to each agent, respectively. The context-dependent distinction was partially neutralised with 4 word swaps and entirely neutralised with 8 or more swaps. On the other hand, Figure 16 shows that the /ɪə, eə/ merger was not affected, or possibly enhanced, by the absence of the context-dependent distinction. This rules out the hypothesis that the predominance of the context-dependent pattern in the original dataset was the cause for the neutralisation of the weaker /ɪə, eə/ pattern.

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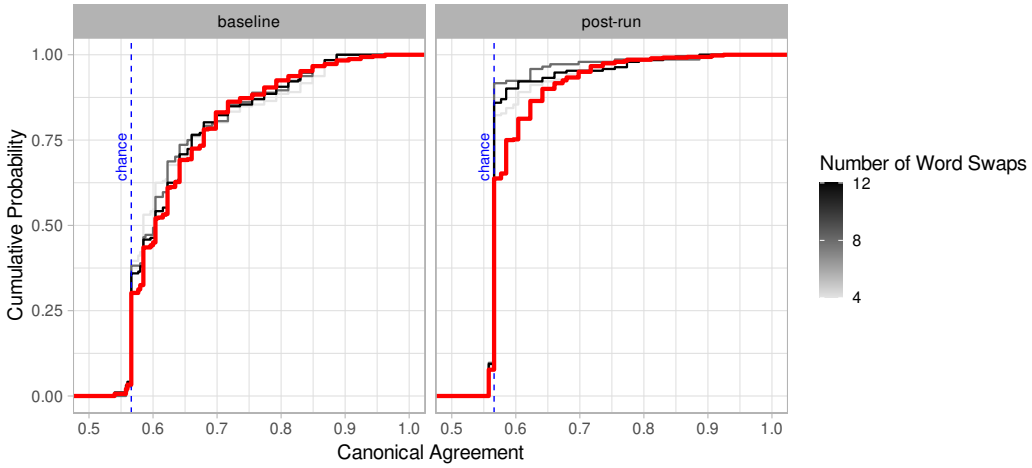

Figure 16: Empirical cumulative distributions of canonical agreement as computed by Eq. (2) at baseline and post-run for 3 batches of 5 ABM simulations. Each batch is based on a manipulated version of the original NZE agent population where a number of word label swaps was carried out (indicated by the gray scale). The swaps were operated in such a way to disrupt the context-dependent pattern described in Table 1 while keeping the canonical /ɪə, eə/ distinction intact. For comparison, the original NZE simulation is indicated by a thicker red curve.

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